

Electromagnetic Induction:

(1)

The phenomenon whereby an emf and hence current is induced in any conductor which is cut across or is cut by a magnetic flux is known as electromagnetic induction.

Interaction: Is the process whereby current carrying conductor placed in a magnetic field experience a force. The force induced by the current carrying conductor is

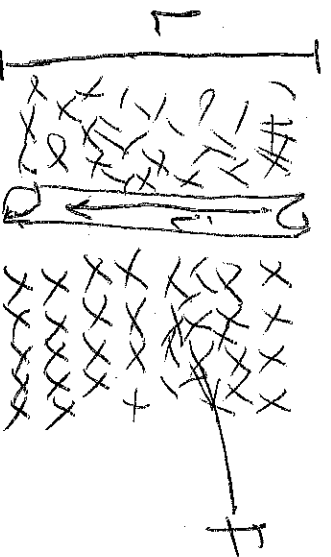
$$F = i(l \times B)$$

Where i is the magnitude of the current in mA
 l - length of the wire in metre (m)
 B - Magnitude of the flux density

The direction of the force is given by Fleming's right hand rule. The magnitude of the force is given by

$$F = i l B \sin \theta$$

Where θ is the angle between the ~~wire~~ ^{coil} and the flux density. The basic concept involved is illustrated in the figure below



Alignment:- Whenever a piece of ferromagnetic material is placed in a magnetic field, a force is experienced upon the material, tending to bring it ~~to~~ into the position of the densest portion of the field. The force tends to align the material specimen so that the reluctance of the magnetic path passing through the material will be at minimum.

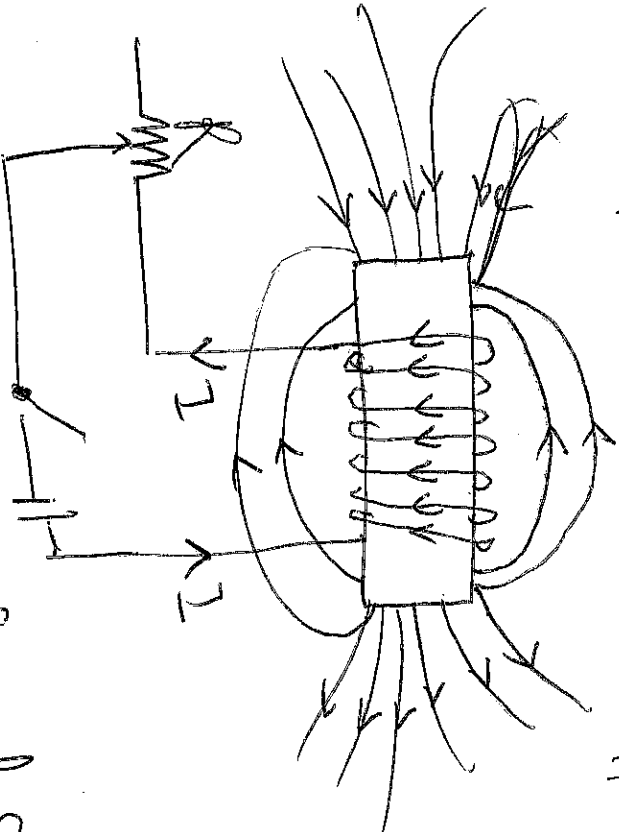
Electromagnetic Induction Continues

(2)

The emf induced by variation of flux is termed as statically induced emf. It is further divided into (1) Self induced emf (2) Mutual induced emf.

Self-induced emf.

Self induced emf is the emf induced in a coil due to the change of its own flux linked with it. If the current through the coil is changed then the flux linked with its own turns will also change which will produce in it, what is called self-induced emf $\mathcal{E} = -\frac{N \frac{d\phi}{dt}}$



Self-induced emf and Self-inductance

The property of the coil due to which it opposes any increase or decrease of current or flux through it is known as self-inductance. Coefficient of self-inductance can be

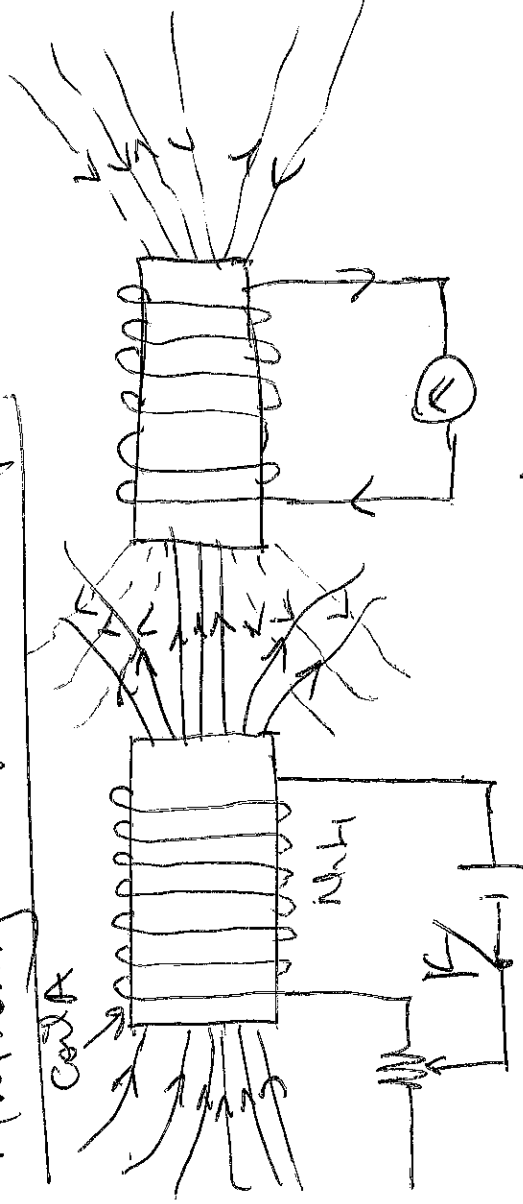
$$(1) \quad L = \frac{N\phi}{I} \quad (H) \quad \text{found using 1-3}$$

where ϕ is the flux

$$\textcircled{2} \quad L = \frac{\mu_0 \mu_r AN^2}{l} \quad \text{CHS}$$

$$\textcircled{3} \quad L = \frac{d\phi}{dI} \quad \text{CHS}$$

Mutually Induced Emf:



Mutually induced emf and Mutual Inductance.

When current through A is established by closing the switch, its magnetic field is set-up and partly links with or threads through the coil B. As the current through A is changed, the flux linked with B is also changed. Hence mutually induced e.m.f is produced in B where magnitude is given by Faraday's law and direction by Lenz's law.

The ability of one coil (or circuit) to produce emf in a nearby coil by induction when the current in the first coil changes is called Mutual Inductance. The action is reciprocal.

(3)

Coefficient of Mutual Induction (M) may be found by the following relations:

$$\textcircled{1} M = \frac{N_2 \Phi_1}{I_1} \text{ (H)}$$

$$\textcircled{2} M = \frac{\mu_0 \mu_r A N_1 N_2}{l} \text{ (HS)}$$

$$\textcircled{3} M = \frac{-\mathcal{E}_m}{\frac{di_1}{dt}}$$

Worked Example: The field winding of a D-C electromagnet is wound with 800 turns and has resistance of 40Ω when exciting voltage is 230 V , the magnetic flux around the coil is 0.004 Wb .

Calculate: $\textcircled{1}$ The self-inductance of the coil
 $\textcircled{ii}$ The energy stored in the magnetic field.

Soln: Given :- $N = 800$, $R = 40 \Omega$, $V = 230 \text{ V}$, $\Phi = 0.004 \text{ Wb}$

$$\textcircled{1} \text{ Self-inductance}$$

$$L = \frac{N \Phi}{I} = \frac{N \Phi}{V/R} = \frac{800 \times 0.004}{230/40}$$

$$= 0.556 \text{ H.}$$

$\textcircled{ii}$ The energy stored in the magnetic field.

$$W = \frac{1}{2} L I^2$$

$$= \frac{1}{2} \times 0.556 \times \left(\frac{230}{40}\right)^2$$

$$= 9.19 \text{ J.}$$

Factors responsible for helminthiasis to be
experienced are as follows;

D.

ELECTRICAL MACHINES IIIELECTRO-MECHANICAL ENERGY CONVERSION.A] Introduction:

Electric Energy ~~can~~ can be transmitted over a long distance with ease and high efficiency; this makes it advantageous over other forms of energy. It is being used as a transmitting link for transporting other form of energy such as Mechanical, Sound, light etc. from one physical location to another.

Therefore, an electromechanical energy conversion device is one which converts electrical energy into mechanical energy and vice versa. It takes place via medium of magnetic or electric field. It is a reversible process.

We have three (3) categories of energy converter.

- 1] Rotating or Linear electrical Machine - e.g. Motors and Generators
- 2] Transducer e.g. Microphone, pictures, Sensors and Loudspeakers
- 3] Force producing device e.g. Solenoid, relays and electro-magnets

B]. Principle of Energy Conversion

The principle of conservation of energy which states that "energy can neither be created nor destroyed, it can merely be converted from one form to another" can be involved in converting energy from one form to another.

In an energy conversion device, out of the total input energy, a larger part of the energy is converted into useful output energy, some energy is stored and rest is converted to heat (this is called energy loss).

C] ENERGY BALANCE.

Electromechanical energy conversion involves energy in four forms, and conservation of energy leads to the following relation among these forms:

$$\left(\begin{array}{l} \text{Energy input} \\ \text{from electrical} \\ \text{source} \end{array} \right) = \left(\begin{array}{l} \text{Mechanical} \\ \text{energy} \\ \text{output} \end{array} \right) + \left(\begin{array}{l} \text{increase in} \\ \text{energy stored} \\ \text{in coupling} \\ \text{field} \end{array} \right) + \left(\begin{array}{l} \text{energy} \\ \text{converted} \\ \text{to heat} \end{array} \right) \quad \text{--- eqn (1)}$$

The equation (1) above is applicable to all conversion devices. The electrical and mechanical terms have a positive value for motor and negative value for generator.

- ① Irreversible conversion of energy to heat arises from three causes:
 - ① Part of the electrical energy is converted directly to heat in the resistances of the current path.
 - ② Part of the mechanical energy developed within the device is absorbed in friction and windage and converted to heat.
 - ③ Part of the energy absorbed by the coupling field is converted to heat in magnetic core loss [for magnetic coupling] or dielectric loss [for electric coupling].

If the energy losses in the electrical system, the mechanical system, and the coupling field are grouped with the corresponding terms in eqn (1), the energy balance may be written in the following form:

$$\left(\begin{array}{l} \text{Electrical energy} \\ \text{input minus} \\ \text{resistance losses} \end{array} \right) = \left(\begin{array}{l} \text{Mechanical energy} \\ \text{output plus} \\ \text{friction and} \\ \text{windage losses} \end{array} \right) + \left(\begin{array}{l} \text{increase in energy} \\ \text{stored in coupling} \\ \text{field plus} \\ \text{associated losses} \end{array} \right) \quad \text{--- eqn (2)}$$

1 below depicts the schematic diagram of equation (2) (2)

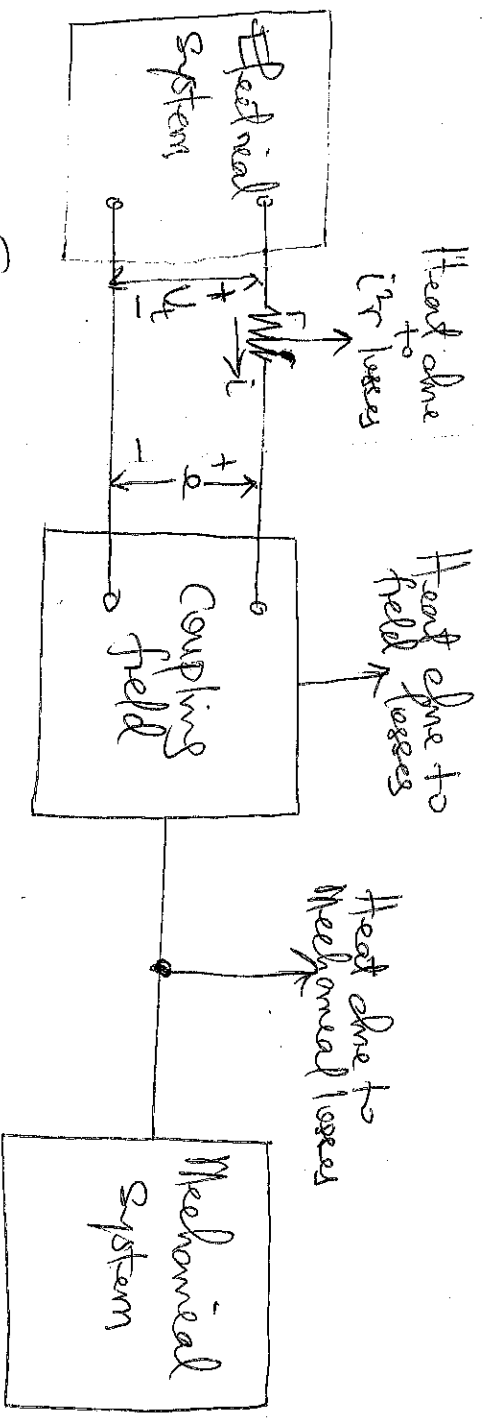


Fig 1: General Representation (Schematic) of electromechanical energy conversion.

Consider fig 1 above,
The differential energy input from electrical source in time dt is $V_i i dt$

where V_i is the instantaneous terminal voltage
 i is the instantaneous current.

The energy loss in the resistance of the device is $i^2 r dt$ where r is the resistance.

Hence left hand side of eq (2) considered

$$dW_{\text{elect}} = V_i i dt - i^2 r dt \quad \text{--- eqn (2)}$$

$$dW_{\text{elect}} = (V_i - ir) i dt \quad \text{--- eqn (4)}$$

Where dW_{elect} is the net electrical energy input to the coupling device after resistance have been taken into account.

For the coupling device to absorb energy from the electric circuit, the coupling field must produce a reaction in the circuit. This reaction is the emf indicated by $\mathcal{E}$ in fig 1.

For voltage to balance properly, the emf, $\mathcal{E}$ must be

$$\mathcal{E} = V_i - ir \quad \text{--- eqn (5)}$$

Substitutes eq (5) into (4) $dW_{\text{elect}} = \mathcal{E} i dt \quad \text{--- (6)}$

Eqn (6) is the total electrical energy input. If electrical energy is supplied to the coupling field from more than one source, the total electrical energy input is the sum of terms of the form of eqn 6.

In differential form eqn 2 may be written

$$dW_{elect} = e i dt = dW_{fld} + dW_{mech} \quad \text{--- (7)}$$

where dW_{mech} is the differential energy converted to mechanical form and dW_{fld} is the differential energy absorbed by the coupling field.

Eqn 7, together with Faraday's law for induced voltage, is the fundamental basis of analysis of energy-conversion devices.

SINGLE EXCITED MAGNETIC FIELD SYSTEM

Whenever electromechanical conversion takes place through only one A-C or D-C source it is known as single-excited magnetic field systems. Examples are electromagnetic relay, reluctance motor etc.

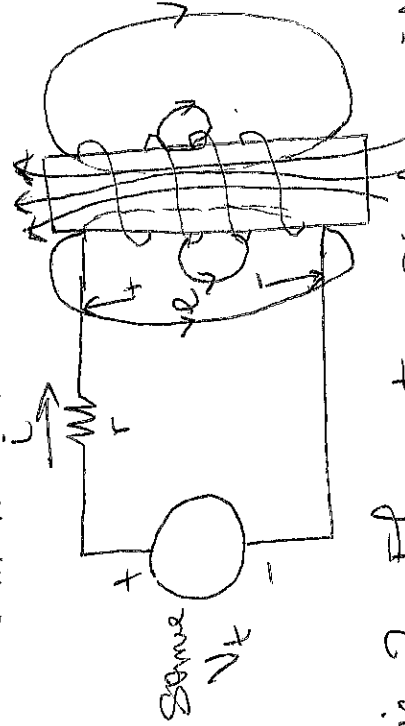


Fig 2. Elementary single-excited Magnetic system.

Consider the fig 2 above comprises of single-excited system

$$V_t = i r + e \quad \text{--- (8)}$$

The emf, e , induced in the coil is in the direction to oppose the current as the flux increases.

$$\therefore e = + \frac{d\lambda}{dt} \quad \text{--- (9)}$$

where λ is the instantaneous value of the flux linkage and t is the time.

For a winding in which all of the flux Φ links all N turns (3) of the winding the flux linkage λ is $N\Phi$.

From eqn (8)

$$V_t = i r + \frac{d\lambda}{dt} = i r + N \frac{d\Phi}{dt} \quad (10)$$

Volume $\Phi = \frac{\lambda}{N}$ or $\lambda = \Phi N$

From eqn (6)

$$dW_{\text{elect}} = e i dt \quad (11)$$

Substitutes eqn (11) into eqn (9)

$$dW_{\text{elect}} = \frac{d\lambda}{dt} \cdot i \cdot dt$$

$$= i d\lambda = i \cdot d\Phi N = N i d\Phi \quad (12)$$

or $dW_{\text{elect}} = F d\Phi$

Volume $F \equiv \text{MMF}$ is the magnetomotive force or MMF of the coil.

Equation (12) gives the electrical energy input.

2. ENERGY IN THE MAGNETIC FIELD.

In the energy balance equation (7), the energy associated with the magnetic field is determined by the MMF of the coil. Mechanical forces are created by the field, but if there is no mechanical motion no mechanical work will be done.

Equation (7) can then be reduced to

$$dW_{\text{elect}} = dW_{\text{fld}} + 0 \quad (13)$$

Substitutes eqn (12) into eqn (13)

$$dW_{\text{elect}} = dW_{\text{fld}} = i d\lambda = F d\Phi \quad (14)$$

i.e. for a fixed configuration, the electrical energy input $i d\lambda$ associated with a change in flux is absorbed by the field. The energy absorbed by the field in changing the flux linkage from λ_1 to λ_2 or the flux from Φ_1 to Φ_2 is

$$\Delta W_{\text{fld}} = \int_{\lambda_1}^{\lambda_2} i(\lambda) d\lambda = \int_{\Phi_1}^{\Phi_2} F(\Phi) d\Phi \quad (15)$$

If the initial flux is zero, the energy absorbed by the field

$$W_{\text{fld}} = \int_0^{\lambda} i(\lambda) d\lambda = \int_0^{\Phi} F(\Phi) d\Phi \quad (15)$$

Electromagnetic energy conversion devices are built with air gaps in the magnetic circuit to separate the fixed and moving parts.

Most of the mmf of the winding is required to overcome the air gap ~~and~~ reluctance so that most of the energy is then stored in the air gap and is returned to the electric source when the field is reversed.

Consider the following relationship

$$\mathcal{L} = \frac{F}{\Phi}, \text{ where } \mathcal{L} \text{ is the reluctance, } F, \text{ mmf and } \Phi \text{ flux} \quad (17)$$

$$\mu = \frac{\Phi}{F} = \frac{1}{\mathcal{L}}, \text{ where } \mu \text{ is the permeance} \quad (18)$$

With non-linearity and hysteresis neglected, the reluctance and permeance are constant with Φ and F .
Integrate eqn 16

$$\begin{aligned} \text{i.e. } W_{fld} &= \int_0^\lambda i(\lambda) d\lambda \\ &= i \left[\frac{\lambda}{2} \right]_0^\lambda \\ &= \frac{1}{2} i \lambda \Rightarrow \frac{1}{2} i \cdot \lambda \end{aligned}$$

$$\int x^n = \frac{x^{n+1}}{n+1}$$

Since $F = iN$, $\lambda = N\Phi$,

$$\text{then } W_{fld} = \frac{1}{2} \cdot \frac{F}{N} \cdot N\Phi$$

$$\begin{aligned} &= \frac{1}{2} F \Phi \\ &= \frac{1}{2} \cdot R \Phi \cdot \Phi \\ &= \frac{1}{2} R \Phi^2 \end{aligned}$$

Therefore, $W_{fld} = \frac{1}{2} i \lambda = \frac{1}{2} F \Phi = \frac{1}{2} R \Phi^2 \quad (19)$
Under non-linearity condition the energy is specified as in eqn (19).

Under the same conditions of linearity, the self-inductance L of the coil in henry is

$$L = \frac{\lambda}{i} = \frac{N\Phi}{i} = \frac{N}{l} \cdot \mu F \approx \frac{N}{l} \cdot \mu \cdot i \quad (19)$$

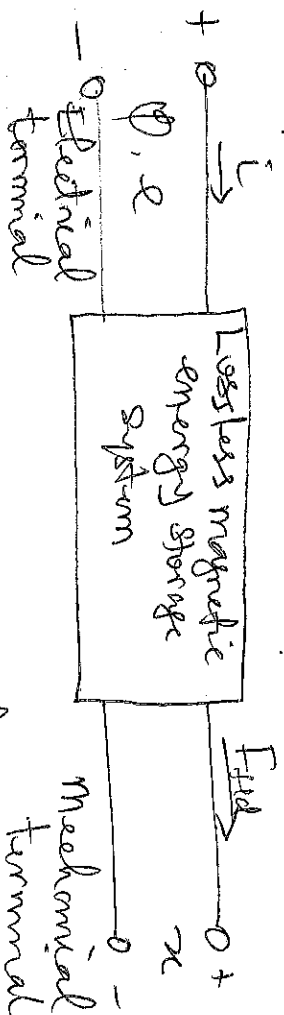
$$\lambda = Li \approx N^2 \mu \quad (20)$$

Substituting $L = \frac{\lambda}{i} \approx i = \frac{\lambda}{L}$ into eqn (19)

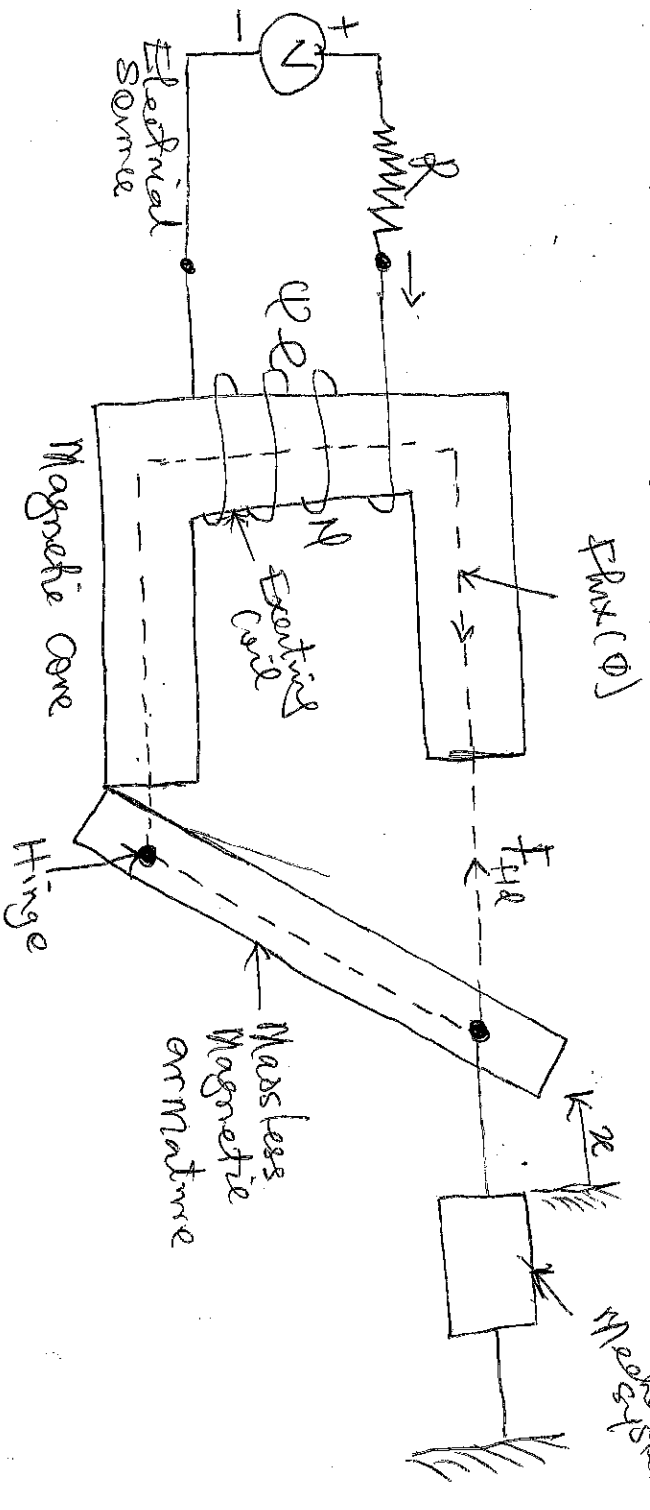
$$W_{fld} = \frac{1}{2} \frac{\lambda^2}{L} \approx \frac{\frac{1}{2} L^2 i^2}{\frac{1}{2} L i^2} = \frac{1}{2} L i^2 \quad (21a)$$

$$= \frac{1}{2} L i^2 \quad (21b)$$

Fig 3. below depict a well fashioned singly-excited magnetic field system.



(a) Schematic Magneto-field electro-mechanical-energy conversion device.



(b) Simple force producing device - attracted armature relay.

Fig 3: Singly-excited Magnetic field System.

WORKED Example

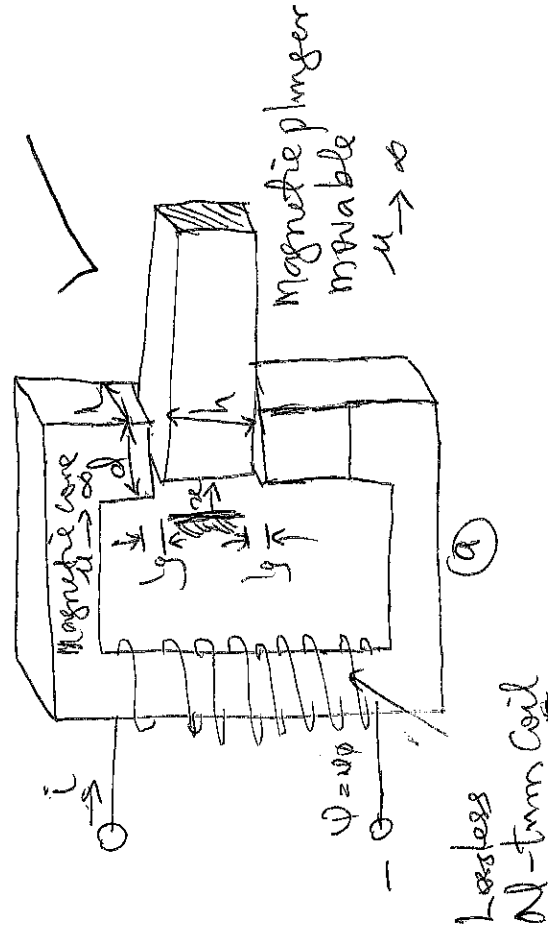
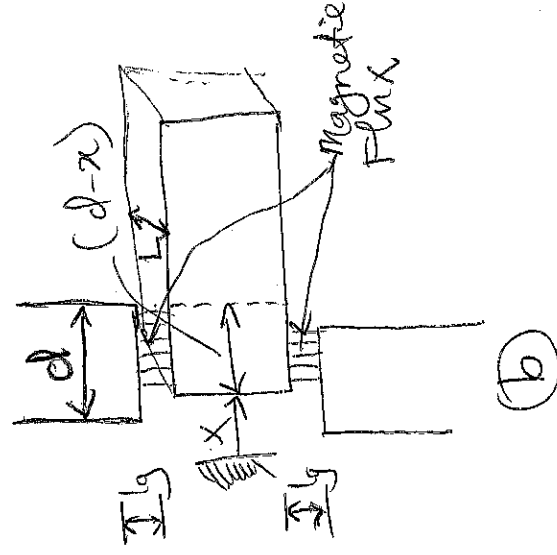


Fig. 1 (a) Relay with Movable plunger

(b) Detail showing air-gap configuration with the plunger partially removed.



Example 1 : Fig 1 shows a relay made from infinitely-permeable magnetic material with a movable plunger, also of infinite-permeable material. The height of the plunger is much greater than the air gap length ($h \gg l_g$). Calculate the Magnetic stored energy W_{fld} as a function of plunger position ($0 < x < d$) for $N = 1200$ turns, $l_g = 1.8 \text{ mm}$, $d = 0.12 \text{ m}$, $l = 0.08 \text{ m}$ and $i = 12 \text{ A}$.

Solution :

$$N = 1200 \text{ turns}; l_g = 1.8 \text{ mm}; d = 0.12 \text{ m}; l = 0.08 \text{ m}; i = 12 \text{ A}$$

Magnetic Stored Energy W_{fld}

Recall: $W_{fld} = \frac{1}{2} L(x) i^2$

Where L is the inductance

3) DYNAMIC EQUATIONS FOR SIMPLY EXCITED ELECTROMECHANICAL SYSTEM. (16)

In the previous section, we have dealt with how the conversion device (model) in electromechanical conversion operates. In this section, we are interested in how the complete electromechanical system operates with the conversion device contained within it.

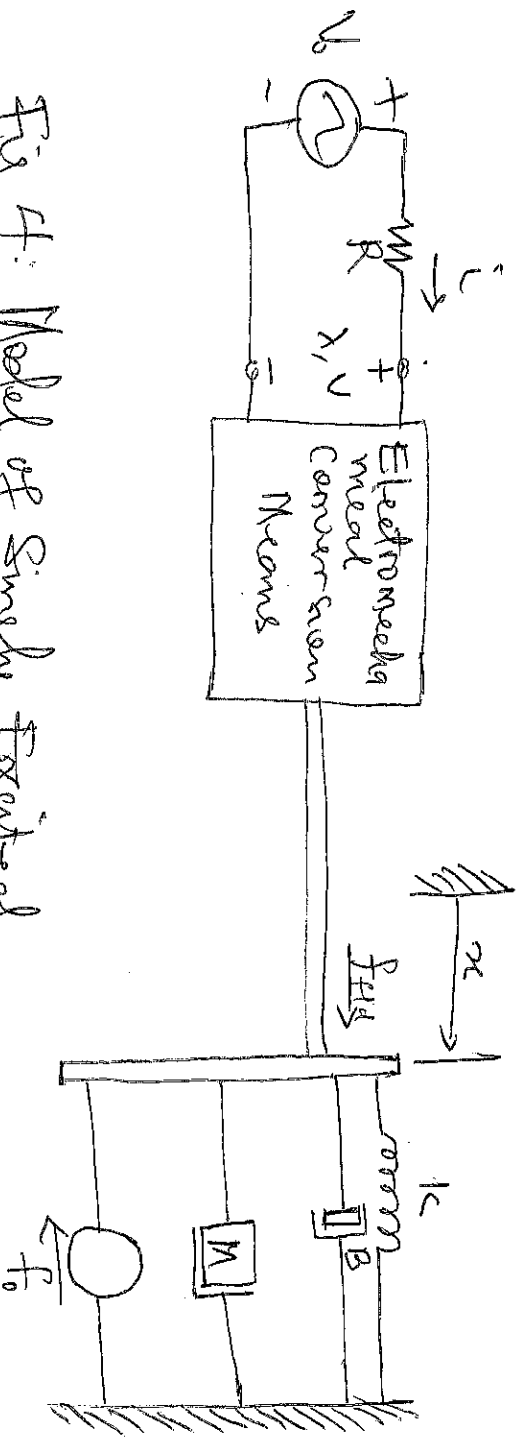


Fig 4: Model of Simply Excited Electromechanical System.

The Model which comprises the basic portions of a Simple Conversion system is shown in Fig 4 above. It consists of three parts M_3 :

- ① An external electrical portion
- ② The energy-conversion portion
- ③ The external Mechanical portion.

The electrical portion is represented by a Voltage Source V_0 and Resistance R . The Source could also be represented by a Current source and a parallel conductance G . The electrical Equation:

$$V_0 = iR + \frac{d\lambda}{dt} \quad \text{--- (22)}$$

but λ (flux linkage) = $L(x)i$
 from the external Equation becomes

$$V_0 = iR + L(x) \frac{di}{dt} + i \frac{dL(x)}{dx} \frac{dx}{dt} \quad \text{--- (23)}$$

The term $L \frac{d^2x}{dt^2}$ is the Self-inductance Voltage, the term $i \frac{dL(x)}{dx} \frac{dx}{dt}$ is the velocity voltage term which is responsible for the energy transfer between the external electrical system and the energy-conversion means.

Mechanical Portion: The Mechanical Portion of Fig 4 shows the symbols for the Spring K , clamping B , and Mass M . These symbols relate force to displacement, velocity and acceleration.

$$f_K = K(x - x_0) \quad \text{--- (24)}$$

$$f_B = B \frac{dx}{dt} \quad \text{--- (25)}$$

$$f_M = M \frac{d^2x}{dt^2} \quad \text{--- (26)}$$

where x_0 is the value of x with the Spring unstretched and applied mechanical force $f_0 = 0$.

The force-balance equation for the mechanical portion is thus:

$$f_{fd} = f_K + f_B + f_M + f_0 \quad \text{--- (27)}$$

$$f_{fd} = K(x - x_0) + B \frac{dx}{dt} + M \frac{d^2x}{dt^2} + f_0 \quad \text{--- (28)}$$

The differential equations for the overall system of Fig 4 for arbitrary inputs $V_{o(t)}$ and $f_0(t)$ are

$$V_{o(t)} = iR + L(x) \frac{di}{dt} + i \frac{dL(x)}{dx} \frac{dx}{dt} \quad \text{--- (29)}$$

$$-f_0(t) = M \frac{d^2x}{dt^2} + B \frac{dx}{dt} + K(x - x_0) - f_{fd}(x, i) \quad \text{--- (30)}$$

The functions $L(x)$ and $f_{fd}(x, i)$ depend upon the properties of the conversion device.

$$L_{crs} = \frac{\mu_o \omega^2 A_g}{2 l_g}$$

where μ_o = absolute permeability = $4\pi \times 10^{-7} \text{ H/m}$

M = number of turns

A_g = gap cross-sectional area

l_g = length of Air gap each, above and below the plunger

Cross-sectional area A_g from the fig 4 above

$$A_g = L(d - 2x)$$

$$= dL \left(1 - \frac{x}{d}\right)$$

Thus

$$L_{crs} = \frac{\mu_o \omega^2 L d \left(1 - \frac{x}{d}\right)}{2 l_g}$$

$$W_{fld} = \frac{1}{2} \left[\frac{\mu_o \omega^2 L d \left(1 - \frac{x}{d}\right)}{2 l_g} \right] l^2$$

$$= \frac{1}{2} \left[\frac{(4\pi \times 10^{-7}) (1200)^2 (0.08) (0.12) \left(1 - \frac{x}{d}\right)}{2 (1.8 \times 10^{-3})} \right] \times 12^2$$

$$= 347.4 \left(1 - \frac{x}{d}\right) \text{ J.}$$



Example 2: The following data relate to the electromagnet relay shown in Fig 3.

Number of turns, $N = 1200$

Cross-Sectional Area of the Core, $A_{core} = 2.7 \text{ cm} \times 2.7 \text{ cm}$

Permeability of the Core = Infinite

Neglecting fringing effects, find the following:

(i) The Inductance of coil for an air gap length (l_g) of 0.45 cm. The energy stored in the field and force acting on the armature when the coil carries a current of 2.25 A

(ii) The mechanical energy output when the air gap is decreased from 0.45 cm to 0.225 cm, assuming that the current through the coil remains constant at 2.25 A

(iii) The expression for force on armature as a function of l_g , coil current being constant at 2.25 A. Also calculate the work done by the magnetic field when l_g changes from 0.45 cm to 0.225 cm from the expression $\int F_{fld} dl_g$ and thereby verify the result of part (ii).

Solution;

(i) L is W_{fld} ; F_{fld} :

$$L = \frac{\mu_0 N^2 A_{core}}{l_g} = \frac{(4\pi \times 10^{-7}) \times (1200)^2 \times (2.7 \times 2.7 \times 10^{-4})}{(0.45 \times 10^{-2})}$$

$$= 0.293 \text{ H}$$

$$W_{fld} = \frac{1}{2} L i^2 = \frac{1}{2} \times 0.293 \times (2.25)^2 = 0.742$$

$$F_{fld} = \frac{W_{fld}}{l_g} = \frac{0.742}{(0.45 \times 10^{-2})} = 164.89 \text{ Nm}$$

$$W_{fld} = \text{Force} \times \text{distance}$$

(iii) l_{mech}

$$\phi_{\text{mech}} \approx \frac{1}{2} \phi_{\text{elect}} = \frac{1}{2} \vec{v}_0 (\psi_2 - \psi_1)$$

$$\Rightarrow \frac{1}{2} \vec{v}_0 (\psi_2 - \psi_1) = \frac{1}{2} \vec{v}_0^2 (L_2 - L_1)$$

$$\cancel{\psi} = \psi \phi$$

(6)

$$A \quad L = \frac{\mu_0 \mu^2 A_{\text{core}}}{l_g} = \frac{(4\pi \times 10^{-7}) \times (1200)^2 (2.7 \times 2.7 \times 10^{-4})}{0.001319 l_g}$$

$$\therefore L_1 (l_g = 0.45 \times 10^{-2} \text{ m}) = \frac{0.001319}{0.45 \times 10^{-2}} = 0.293 \text{ H}$$

$$L_2 (l_g = 0.225 \times 10^{-2} \text{ m}) = \frac{0.001319}{0.225 \times 10^{-2}} = 0.586 \text{ H}$$

Substituting the values, we have

$$l_{\text{mech}} = \frac{1}{2} \times (2.25)^2 (0.586 - 0.293) = 0.742 \text{ S.}$$

(iv) Expression for force (F_{fld}) as a function of l_g and l_{mech}

$$F_{\text{fld}} = \frac{W_{\text{fld}}}{l_g} = -\frac{1}{2} \frac{L \psi^2}{l_g} = -\frac{1}{2} \left[\frac{\mu_0 \mu^2 A_{\text{core}}}{l_g \times l_g} \right] \psi^2$$

Negative sign indicates that force acts in a direction to reduce air gap length (l_g).

$$= -\frac{1}{2} \left[\frac{4\pi \times 10^{-7} \times 1200^2 \times (2.7 \times 2.7 \times 10^{-4})}{l_g^2} \right] \times (2.25)^2$$

$$\therefore F_{\text{fld}} = -\frac{0.00334}{l_g^2} \quad \text{--- Required expression.}$$

$$\begin{aligned}
 \Delta W_{\text{mech}} &= \int_{g_1}^{g_2} F_{\text{fd}} \, dg \\
 &= \int_{0.45 \times 10^{-2}}^{0.225 \times 10^{-2}} \frac{0.00334}{g^2} \, g^2 \\
 &= 0.00334 \left[\frac{1}{0.225 \times 10^{-2}} - \frac{1}{0.45 \times 10^{-2}} \right] \\
 &= 0.742 \text{ J}
 \end{aligned}$$

$$\oint \frac{v}{n+1} \, dn$$

Determination of Magnetic or Mechanical force.

①

Recall

$$W_{fld} = \frac{1}{2} L_{ms} i^2$$

$$W_{fld} i_{ms} = \frac{1}{2} L_{ms} i^2$$

$$dW_{fld} = \frac{1}{2} i^2 dL_{ms}$$

The Mechanical force developed then =

$$F_{fld} = \frac{dW_{fld}}{dx} = \frac{1}{2} i^2 \frac{dL_{ms}}{dx}$$

The above equation shows that the Mechanical or Magnetic force developed acts in a direction to increase the exciting coil inductance.

Hence, $W_{fld} = \frac{1}{2} \frac{\lambda^2}{L_{ms}}$

$$W_{fld}(\lambda, x) = \frac{1}{2} \frac{\lambda^2}{L_{ms}}$$

So $F_{fld} = -\frac{dW_{fld}(\lambda, x)}{dx} = \frac{1}{2} \left[\frac{\lambda^2}{L_{ms}} \right]^2 \frac{dL_{ms}}{dx}$

Since $i = \lambda / L$

Also, $W_{fld} = \frac{1}{2} S \Phi^2$

$$F_{fld} = \frac{-dW_{fld}}{dx} = -\frac{1}{2} \Phi^2 \frac{dS}{dx}$$

* How is force between two parallel faces in a ~~single~~ ⁽²⁾ ~~single~~ excited system calculated?

→ Force between two parallel faces in a single excited system

Let the cross-sectional ^{of the magnetic surface} area be A in m^2 , distance between the two faces $= g$, For sharp corner at x in the direction of force F_{fld} , distance b/w the two surfaces is reduced to $(g-x)$

Force b/w parallel faces is given as

$$F_{fld} = -\frac{1}{2} \Phi_g^2 \frac{dS_g}{dx} \quad \text{--- (1)}$$

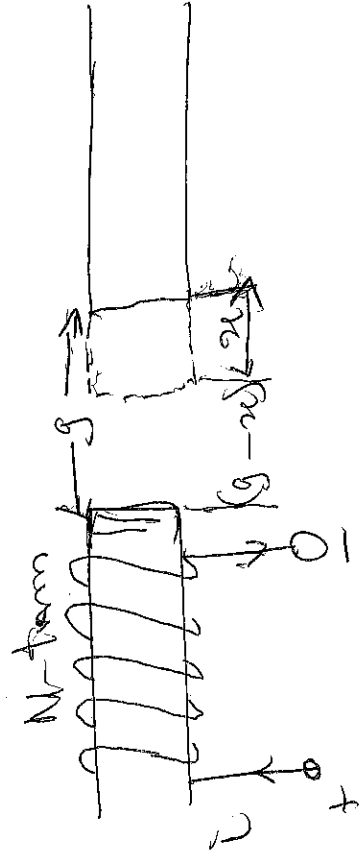
Reluctance of gap length, $S_g = \frac{g-x}{\mu_0 A}$ --- (11)

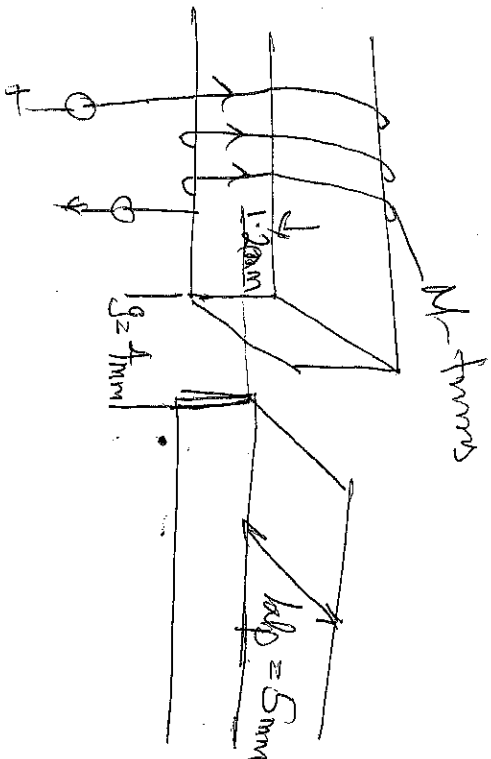
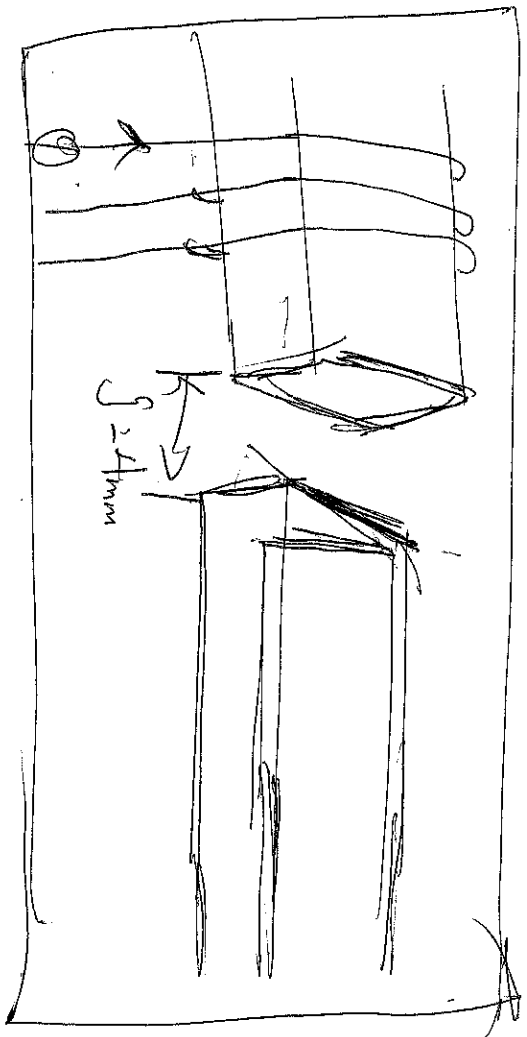
Substituting (11) into (1)

$$\begin{aligned} F_{fld} &= -\frac{1}{2} \Phi_g^2 \frac{d}{dx} \frac{(g-x)}{\mu_0 A} \\ &= -\frac{1}{2} \Phi_g^2 \times \frac{1}{\mu_0 A} \times (-1) \\ &= \frac{1}{2} (B_g A)^2 \times \frac{1}{\mu_0 A} \\ &= \frac{B_g^2 A}{2 \mu_0} \end{aligned}$$



Where $\Phi_g = \text{flux}$
 $B_g = \text{flux density}$





A movable part of an electromagnet is displaced axially as shown in the diagram above. Pole width is 5 mm, flux density is 1.2 T and gap length is 4 mm. If the gap length is kept constant, determine the force that acts in a direction to bring them into axial alignment.

Soln

Reluctance of the overlapping region (S_g)

$$S_g = \frac{g}{\mu_0 A} = \frac{g}{\mu_0 x_{mp}} \quad ; \quad A = x_{mp}$$

$$W_{fel} = \frac{1}{2} S_g \Phi_g^2 = \frac{1}{2} \frac{g}{\mu_0 x_{mp}} \Phi_g^2$$

$$F_{fa} = - \frac{\partial W_{fel}}{\partial x} = - \frac{1}{2} \frac{g \Phi_g^2}{\mu_0 x_{mp}} \frac{\partial}{\partial x} \left(\frac{1}{x} \right)$$

$$= \frac{1}{2} \Phi_g^2 \frac{g}{\mu_0 x_{mp} x^2}$$

$$\frac{\frac{1}{x} x^{-1}}{x^{-1-1}} = \frac{x^{-2}}{x^{-2}} = 1$$

Substituting $\Phi_g = B_g W_p x \left[B_g = \frac{\Phi_g}{A} \right]$

$$F_{fd} = \frac{1}{2} (B_g W_p x)^2 \times \frac{g}{\mu_0 W_p x^2}$$

$$= \frac{1}{2} \frac{B_g^2 W_p g}{\mu_0}$$

Substituting $B_g = 1.116 \text{ T}$ (or Tesla), $W_p = 5 \text{ mm}$ or 0.005 m
 $g = 4 \text{ mm}$ or 0.004 m and $\mu_0 = 4\pi \times 10^{-7}$ in the above
 expr. &

$$F_{fd} = \frac{1}{2} \times (1.1^2 \times 10^{-5}) (0.004)$$

$$= \underline{\underline{7.96 \text{ mN}}}$$

Question

Forces due to
 magnetic field & electric field

$$I_0 = I_m + I_e$$

$$I_m = I_0 - I_e$$

$$\frac{I_0}{-2+1} = \frac{I_e}{-1}$$

$$x = 1$$

✓

Example.

①

A 3-phase step-down transformer is connected to 6.6 kV main and takes 10A. Calculate the Secondary line voltage, line current and output for the following connections. The ratio of turns per phase is 12.

① Δ/Δ (ii) Y/Y (iii) Δ/Y (iv) Y/Δ

Soln

Given

Primary line voltage, $V_{L1} = 6.6 \text{ kV} = 6,600 \text{ V}$

Primary line current, $I_{L1} = 10 \text{ A}$

$$\text{Turn Ratio} = \frac{N_1}{N_2} = 12$$

$$K = \frac{N_2}{N_1}$$

① Δ/Δ connection

Primary phase voltage, $V_{P1} = V_{L1} = 6,600 \text{ V}$

Secondary line voltage, $V_{L2} = V_{P2} = V_{P1} \times \frac{N_2}{N_1}$

$$= \frac{6,600}{12} = 550 \text{ Volts}$$

$$\text{Primary Phase Current, } I_P = \frac{I_{L1}}{\sqrt{3}}$$

$$= \frac{10}{\sqrt{3}} = 5.773 \text{ A}$$

$$\text{Secondary line current, } I_{L2} = \sqrt{3} I_{P2}$$

Output in kVA

$$= \sqrt{3} \times I_P \times \frac{N_1}{N_2}$$

$$\sqrt{3} \times 10 \times 550$$

$$= \sqrt{3} \times 5.773 \times 12$$

$$= 1143.15.35 \approx 114.3 \text{ kVA}$$

$$= 120 \text{ A}$$

(ii) Y/Y Connection

$$\text{Primary Phase Voltage, } V_{P1} = \frac{V_{L1}}{\sqrt{3}} = \frac{6,600}{\sqrt{3}}$$

$$\begin{aligned} \text{Secondary Line Voltage, } V_{L2} &= \sqrt{3} V_{P2} \\ &= \sqrt{3} V_{P1} \times \frac{N_2}{N_1} \\ &= \sqrt{3} \times \frac{6,600}{\sqrt{3}} \times \frac{1}{12} \\ &= \underline{\underline{550V}} \end{aligned}$$

$$\text{Primary Phase Current, } I_{P1} = I_{L1} = 10A$$

$$\text{Secondary Line Current, } I_{L2} = I_{P2} = I_{P1} \frac{N_1}{N_2}$$

$$= 10 \times 12 = 120A$$

(iii) Δ/Y Connections

$$\text{Primary Phase Voltage, } V_{P1} = V_{L1} = 6,600V$$

$$\begin{aligned} \text{Secondary Line Voltage, } V_{L2} &= \sqrt{3} V_{P2} = \sqrt{3} \times V_{P1} \times \frac{N_2}{N_1} \\ &= \sqrt{3} \times \frac{6,600}{12} \end{aligned}$$

$$\begin{aligned} \text{Primary Phase Current } I_{P1} &= \frac{I_{L1}}{\sqrt{3}} \\ &= \frac{10}{\sqrt{3}} = 5.77A \end{aligned}$$

$$\text{Secondary Line Current } I_{L2} = 10/\sqrt{3} = 5.77A$$

$$I_{L2} = I_{P2} = I_{P1} \times \frac{N_1}{N_2} = \frac{10}{\sqrt{3}} \times 12$$

$$= 69.28A$$

$$\begin{aligned} \text{Output in kVA} &= \sqrt{3} \times V_{L2} \times I_{L2} \\ &= \sqrt{3} \times 550 \times 69.28 \end{aligned}$$

$$= \underline{\underline{114.3 kVA}}$$

(2)

(12) Δ connection

$$\text{Primary Phase Voltage, } V_P = \frac{V_{L1}}{\sqrt{3}} = \frac{6,600}{\sqrt{3}} \text{ V}$$

$$\begin{aligned} \text{Secondary line Voltage } V_{L2} &= \sqrt{3} V_P \times \frac{N_2}{N_1} \\ &= \frac{6,600}{\sqrt{3}} \times \frac{1}{12} \\ &= \underline{\underline{317.5 \text{ V}}} \end{aligned}$$

$$\text{Primary Phase Current, } I_P = I_{L1} = 10 \text{ A}$$

$$\text{Secondary line Current, } I_{L2} = \sqrt{3} I_P$$

$$= \sqrt{3} \times I_P \times \frac{N_1}{N_2}$$

$$\begin{aligned} &\approx \sqrt{3} \times 10 \times 12 \\ &= 207.85 \text{ A} \end{aligned}$$

$$\begin{aligned} \text{Output } P &= \frac{\sqrt{3} \times V_{L2} \times I_{L2}}{1,000} = \frac{\sqrt{3} \times 6,600 \times 10}{1,000} \\ &= 114.32 \text{ kVA} \end{aligned}$$

Output I_a kA

$$\begin{aligned} &\frac{\sqrt{3} \times 317.5 \times 207.85}{1,000} \\ &\approx 114.802 \text{ VA} \\ &= \underline{\underline{114.3 \text{ kVA}}} \end{aligned}$$

 Δ

$$\begin{aligned} \text{--- Line Voltage} &= V_P \times \frac{N_2}{N_1} \\ \text{--- Line Current} &= \sqrt{3} \times I_P \times \frac{N_1}{N_2} \\ \text{--- Phase Current} &= \frac{I_L}{\sqrt{3}} \end{aligned}$$

$$\text{Phase voltage} = \frac{\text{Line voltage}}{\sqrt{3}}$$

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Union

Star

$$\text{--- Line voltage} = \sqrt{3} V_P \times \frac{N_2}{N_1}$$

$$\text{--- Line Current} = I_P \times \frac{N_1}{N_2}$$

$$\text{--- Phase voltage} = \frac{V_L}{\sqrt{3}}$$

$$\text{--- Phase Current} = I_P = I_{L1}$$

TRANSFORMER

Introduction

Transformer is an a-c machine that is transfer electrical energy from one electric circuit to another without a change of frequency in which the principle of electromagnetic induction were employed and has electric circuit that are linked by a common magnetic circuit. It is called static transformer because its basic construction requires no moving parts.

2-1 OPERATING PRINCIPLE OF TRANSFORMER.

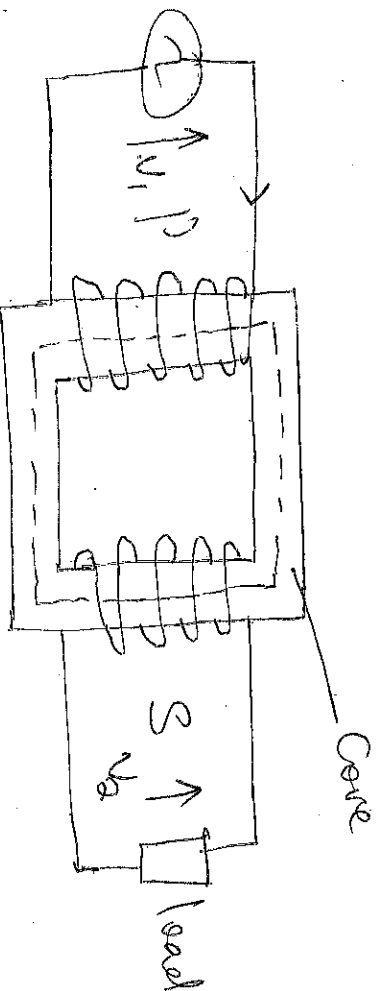


Fig 2-1 Simple Transformer.

A simple transformer as shown in Fig 2-1 consists of a soft iron or ~~steel~~ Silicon Steel core and two winding placed on it. The winding are insulated from both the core and each other. The core provide a path of low reluctance to the magnetic flux.

The winding connected to the supply main is called primary winding while the one connected to the load circuit is secondary winding.

When the primary winding is connected to an a-c supply mains, a current flows through it. Since this winding linked with an iron core, so current flowing through the winding produces an alternating flux ϕ in the core. Since this flux is alternating and links with the secondary winding also, so induces an e.m.f in the secondary winding.

The frequency of induced e.m.f in the secondary is the same as that of the flux or that of the supply voltage. The

Induced emf in the secondary winding enables it to deliver current to an external load connected across it. Thus, the energy is transformed from primary winding to the secondary winding by means of electro-magnetic induction without any change in frequency.

3] E.M.F EQUATION OF A TRANSFORMER:

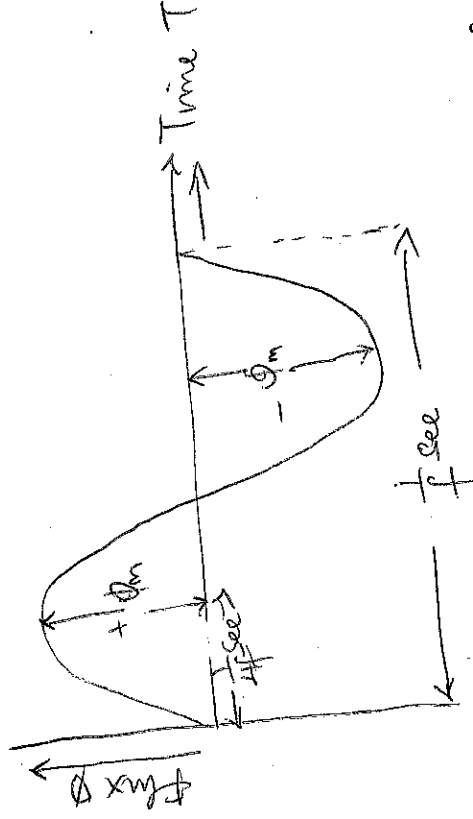


Fig 2-2 Sinusoidal Variation of flux with Time.

When an alternating current (voltage) is applied to the primary winding of a transformer, a sinusoidal alternating current is set-up as shown in Fig 2-2.

Let Φ_{max} = Maximum value of flux, N_1 = No. of turns of primary winding
 f = Supply frequency N_2 = No. of turns of secondary winding.

As shown in Fig 2-2, the magnetic flux increases from zero to maximum value Φ_{max} in one-fourth of a cycle i.e. $T = \frac{1}{4f}$ sec or $\frac{T}{4}$

$$\text{Average rate of change of flux} = \frac{d\Phi}{dt} = \frac{\Phi_{max}}{1/4f}$$

$$= 4f\Phi_{max} \quad \text{--- (1)}$$

Since, average emf induced per turn = Average rate of change of flux

$$\therefore \text{Average emf induced} = 4f\Phi_{max}$$

Hint: Form factor of sinusoidal wave = 1.11.

$$\therefore \text{RMS value of emf induced per turn} = 1.11 \times 4f\Phi_{max} \\ = 4.44f\Phi_{max} \quad \text{--- (2)}$$

RMS value of emf induced in Primary, $E_1 = \text{Emf induced Per turn} \times \text{number of primary turn}$ (2)

$$= 4.44 f \Phi_{\max} \times N_1$$

$$= 4.44 f \Phi_{\max} N_1 \quad \text{--- (3)}$$

Similarly rms value induced in the Secondary $E_2 = 4.44 f \Phi_{\max} N_2 \quad \text{--- (4)}$

Worked Example: OR

RMS value of emf induced in primary $= 4.44 f N_1 B_{\max} A$
Similarly in Secondary $= 4.44 f N_2 B_{\max} A$

where $B_{\max}$ is the maximum flux density
 A is the cross-sectional Area

$$\therefore \Phi_{\max} = B_{\max} A.$$

Worked Example: 1

The emf per turn of a single phase 10kVA, 2200/220V, 50Hz transformer is 10V. Calculate (i) the number of primary and secondary turns (ii) the net cross-sectional area of core for a maximum flux density of 1.5 T.

Soln

$$\text{Emf per turn} = 10 \text{ V}$$

$$\text{Primary induced emf } E_1 = V_1 = 21200 \text{ V}$$

$$\text{Secondary induced emf } E_2 = V_2 = 2200 \text{ V}$$

$$\text{Supply frequency, } f, = 50 \text{ Hz}$$

$$\text{Maximum flux density, } B_{\max} = 1.5 \text{ T}$$

$$(i) \text{ Number of primary turns, } N_1 = \frac{E_1}{\text{Emf per turn}} = \frac{21200}{10}$$

$$\text{Number of secondary turns } N_2 = \frac{E_2}{\text{Emf per turn}} = 220$$

$$= \frac{220}{10} = 22.$$

$$\begin{aligned} \text{Maximum value of flux, } \Phi_{\max} &= \frac{E_{\text{mf per turn}}}{4.44f} \\ &= \frac{10}{4.44 \times 50} = 0.045 \text{ wb.} \end{aligned}$$

(ii) Net cross-sectional area of core

$$A = \frac{\Phi_{\max}}{B_{\max}} = \frac{0.045}{1.5} = 0.03 \text{ m}^2.$$

Worked Example 2

A single phase transformer has 400 primary and 1000 secondary turns. The net cross-sectional area of the core is 60 cm^2 . If the primary winding be connected to a 50 Hz supply at 500 V . Calculate (i) the Peak value of the flux density in the core (ii) the voltage induced in the secondary winding.

Soln

Primary induced emf, $E_1 = V_1 = 500 \text{ V}$

Supply frequency, $f = 50 \text{ Hz}$

Primary turns, $N_1 = 400$

Secondary turns, $N_2 = 1000$

Net cross-sectional area $A = 60 \text{ cm}^2 = 0.006 \text{ m}^2$

(i) Maximum value of flux, $\Phi_{\max} = \frac{E_1}{4.44f}$

$$= \frac{500}{4.44 \times 50} = 0.00563 \text{ wb}$$

(ii) Peak value of flux density in the core, $B_{\max} = \frac{\Phi_{\max}}{A}$

$$= \frac{0.00563}{0.006} = 0.9384 \text{ T}$$

(iii) Voltage induced in the secondary

$$E_2 = E_1 \times \frac{N_2}{N_1} = 500 \times \frac{1000}{400}$$

$$= \underline{\underline{1250 \text{ V}}}$$

A. TRANSFORMER ON NO-LOAD

(3)

When the primary of a transformer is connected to the source of ac supply and the secondary is open, the Xformer is said to be at no-load. i.e. the secondary is not connected to any load.

When an alternating voltage V_1 is applied to the primary winding it will cause the flow of alternating current in the primary winding. Since the primary coil is purely inductive and there is no output (secondary being open), the primary draws the magnetising current ~~only~~ ^{I_m} to magnetize the core. The magnitude of I_m should be zero (ideal Xformer) but the reluctance of the magnetic circuit is never zero therefore I_m has definite magnitude. The magnetising current I_m produces an alternating flux Φ which is at all time proportional to the current and hence in phase with it. The magnetising current I_m is small in magnitude and lags behind supply voltage V_1 by 90° .

Let the instantaneous linking flux Φ be given as

$$\Phi = \Phi_{\max} \sin \omega t$$

Instantaneous values of induced emfs in the primary and secondary winding will be

$$e_1 = N_1 \frac{d\Phi}{dt} = N_1 \frac{d}{dt} (\Phi_{\max} \sin \omega t)$$

$$= N_1 \omega \Phi_{\max} \cos \omega t$$

$$= N_1 \omega \Phi_{\max} \sin \left(\omega t - \frac{\pi}{2} \right)$$

Similarly

$$e_2 = N_2 \omega \Phi_{\max} \sin \left(\omega t - \frac{\pi}{2} \right)$$

Since primary winding has no resistance, therefore applied voltage to primary winding is to only oppose the induced emf in the primary winding hence instantaneous applied voltage to primary will be given by $V_1 = -e_1 = -N_1 \omega \Phi_{\max} \sin \left(\omega t - \frac{\pi}{2} \right)$.

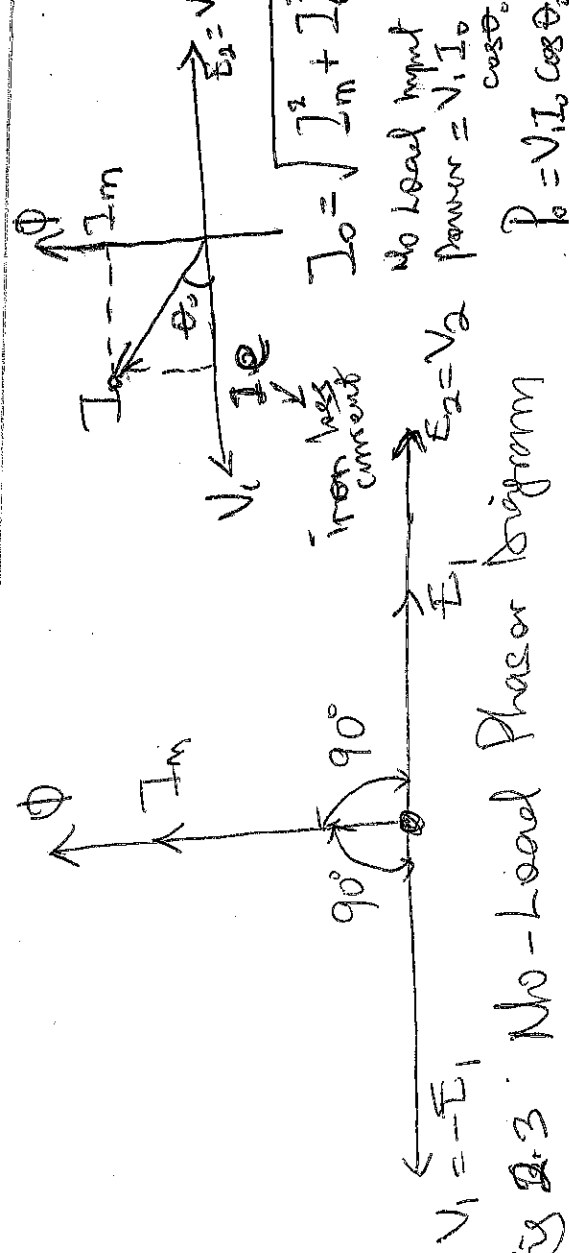


Fig 2.3 No-Load Phasor Diagram

Conclusion

From fig 2.3, it can be concluded as follows:

- ① Induced emfs in primary and secondary windings E_1 & E_2 lag behind the main flux Φ by $\frac{\pi}{2}$, so these emfs (E_1 & E_2) are in phase with each other.
- ② Applied voltage to the primary winding leads the main flux by $\frac{\pi}{2}$ and is in phase opposition to the induced emf in the primary winding.
- ③ Secondary Voltage $V_2 = E_2$ as there is no voltage drop in secondary.
- ④ The magnetising current I_m is proportional to the flux and is in phase with it.

5] TRANSFORMER ON LOAD

When an impedance or load is connected across the secondary terminal of a transformer, it is said to be loaded.

The magnitude and phase of secondary current I_2 with respect to secondary terminal voltage, V_2 will be in phase when the load is non-inductive, lag behind when it is an inductive load and lead the terminal voltage when it is capacitive load.

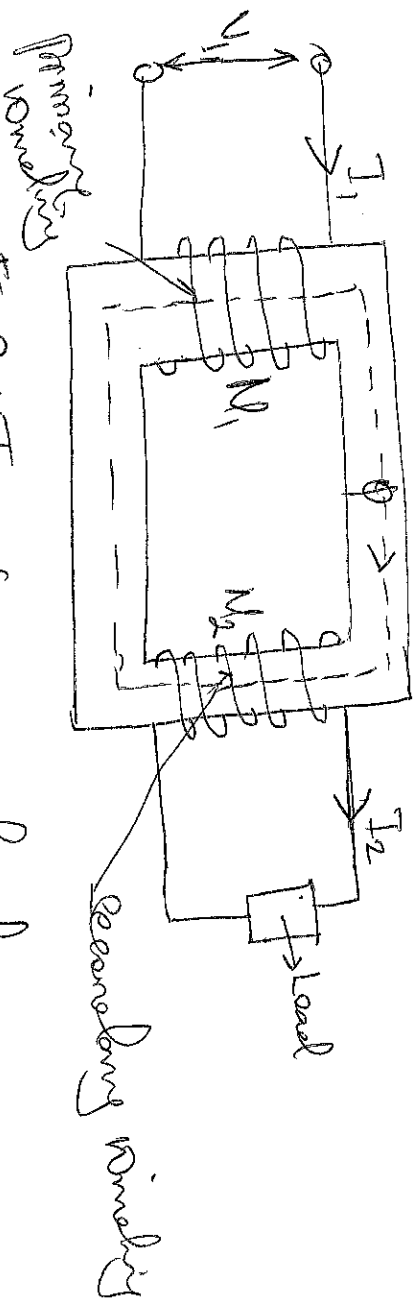
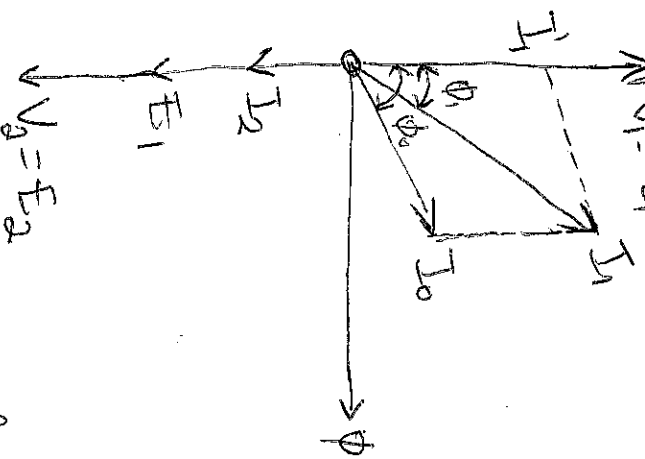


Fig 2.4 Transformer on load

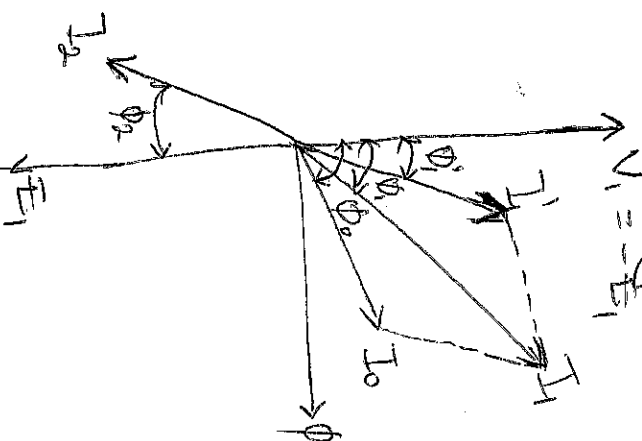
When an impedance is connected across the secondary terminal as shown in Fig 2-4, a current I_2 flows through the secondary winding and set-up its own MMF and hence creates a secondary flux Φ_2 . This secondary flux Φ_2 opposes the main flux Φ set up by the exciting current I_0 . The opposing secondary flux needs the main flux Φ and thereby reduced the primary back-emf. When the applied voltage V_1 increases the back-emf tends to increase and more current is drawn from the source of supply flowing through the primary winding until the original value of flux Φ is obtained.

The phasor diagrams for transformer on non-inductive, inductive and capacitive load is as shown in Fig 2.5.

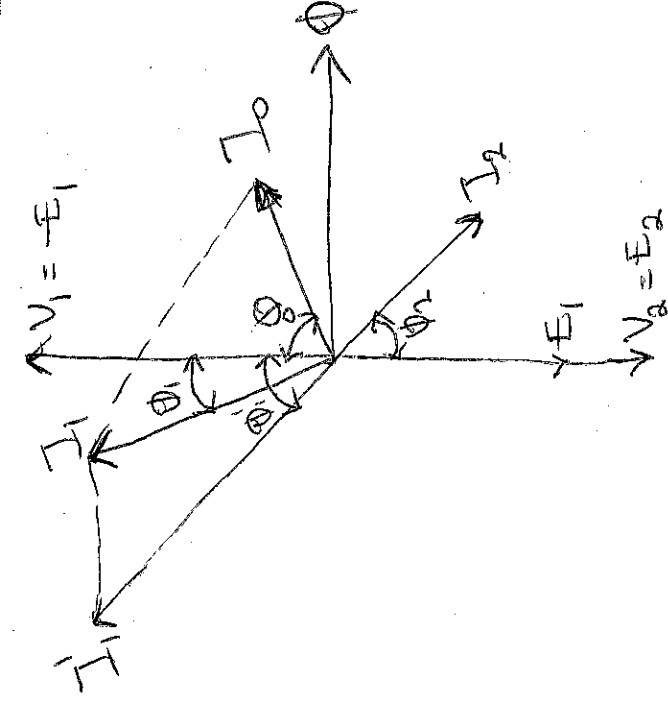
I_0 = Input current or exciting current



① For Pure - Resistive load



② For Inductive load



© For Capacitive load

Fig 2.6 Phasor Diagram for Transformer on load.

The Secondary Current I_2 is in phase, lags behind and leads the Secondary terminal voltage V_2 by an angle θ_2 for pure resistive, inductive and Capacitive loads respectively. The inlead primary current I_1 is also known as Counterbalancing Current and is always in opposition to the Secondary Current I_2 .

I_2 in phase with terminal voltage V_2 [Resistive load]

I_2 lags behind V_2 [Inductive load]

I_2 leads V_2 [Capacitive load]

TRANSFORMER EQUIVALENT CIRCUIT

(5)

An equivalent circuit is merely a circuit interpretation of the equations which describe the behaviour of the device. It is convenient to represent the transformer by an equivalent circuit them by the use of phasor diagram.

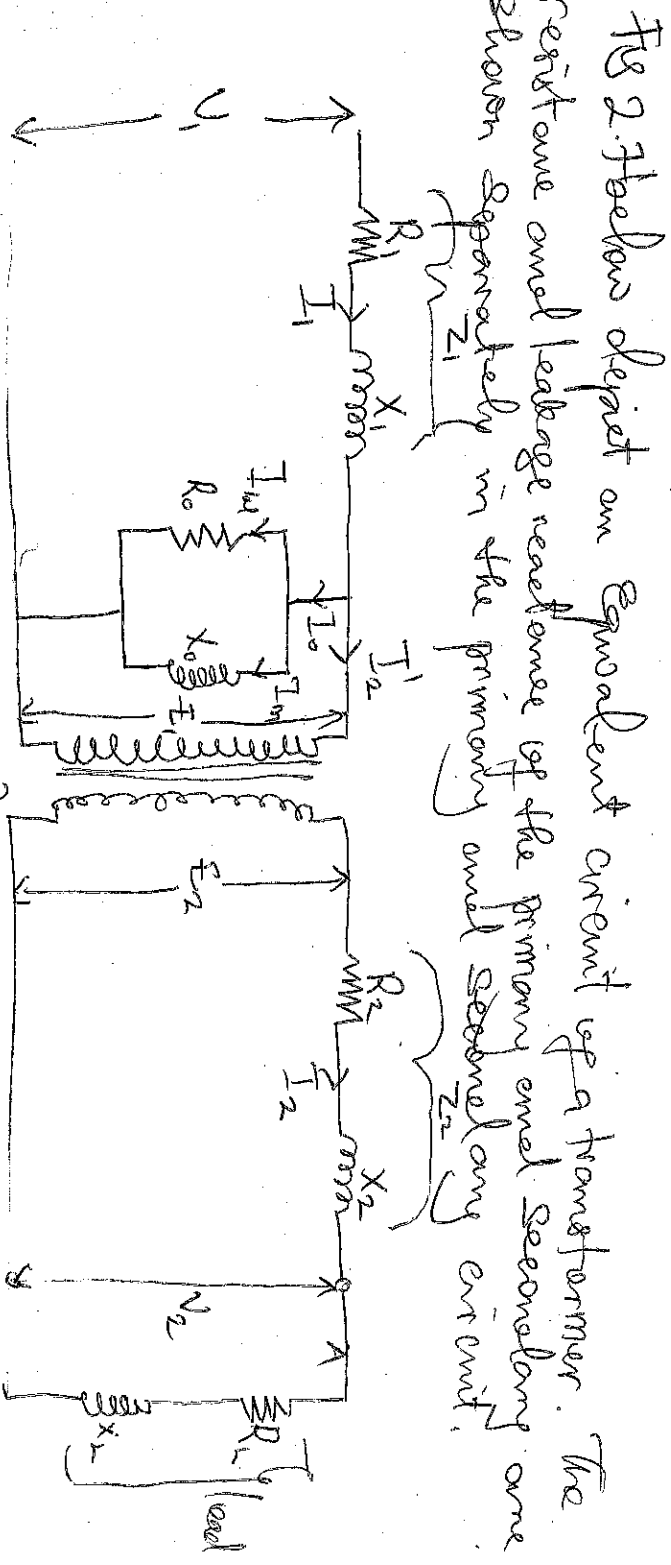


Fig 2.7 Equivalent circuit of a transformer.

The effect of magnetizing current is represented by X_0 . While that of core loss is R_0 .

The equivalent circuit can be simplified by transferring the secondary resistances and reactances to the primary side. Let R_2 = resistance which must be placed in the primary circuit to produce the same drop as produced by R_2 .

~~Then voltage drop across R_2 is $I_2 R_2$~~
 Then voltage drop across by R_2' in primary = $I_2' R_2'$

Ratio of $I_2' R_2'$ and $I_2 R_2$ = turn ratio $\frac{N_1}{N_2}$

$$\text{i.e. } \frac{I_2' R_2'}{I_2 R_2} = \frac{N_1}{N_2}$$

$$\frac{I_2' R_2'}{I_2 R_2} = \frac{N_1}{N_2} = \frac{1}{K}$$

where K is the transformation ratio.

or $R_2 = R_2 \times \frac{I_2}{I_1} \times \frac{1}{K}$

But $\frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{K}$

$K = \frac{N_2}{N_1}$

Substitutes eqn * into eqn **

$R_2' = \frac{R_2}{K^2}$

Similarly $X_2' = \frac{X_2}{K^2}$

from eqn * $I_2' = KI_2$

The exact equivalent circuit of the transformer with all impedances transferred to the primary side is shown in Fig. 2.8.

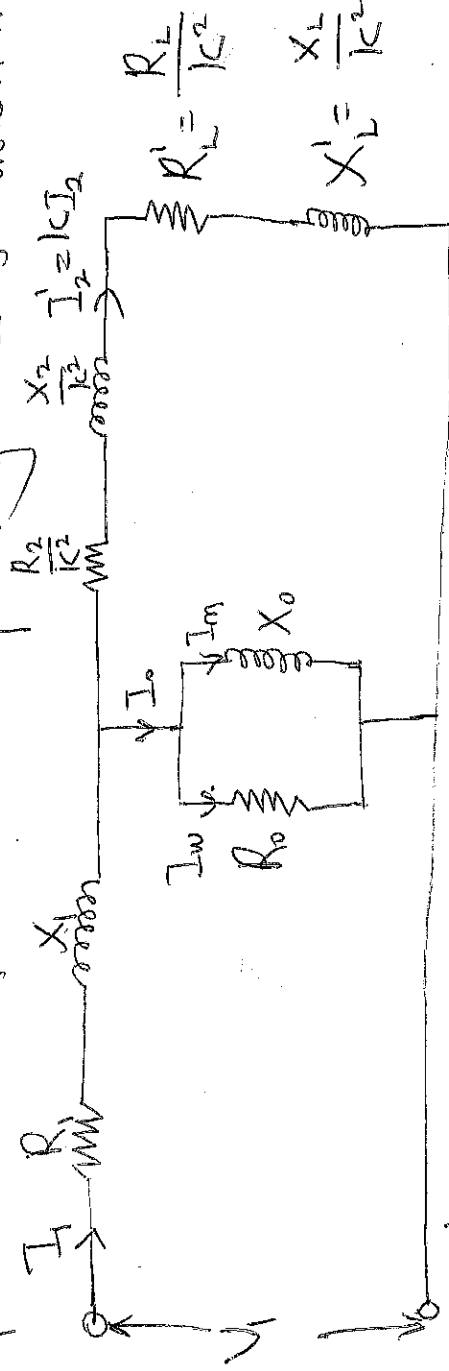


Fig. 2.8. Equivalent Circuit with Secondary Impedances transferred to Primary.

Note: When primary winding resistance R_1 and reactance X_1 are transferred to secondary, it can be denoted by R_1' and X_1' and are given as

$R_1' = K^2 R_1$
 $X_1' = K^2 X_1$

APPROXIMATE EQUIVALENT CIRCUIT OF TRANSFORMER. (6)

Fig 2-9 shows the approximate equivalent circuit of transformer. The equivalent circuit can be simplified by transferring the parallel branch consisting of R_o & X_o to the extreme left portion as shown in Fig 2-9. This circuit is known as approximate equivalent circuit. The advantage of this circuit is that simple to analyse because the Resistance R_1 and R_2 and leakage reactance X_1 and X_2 can be combined.

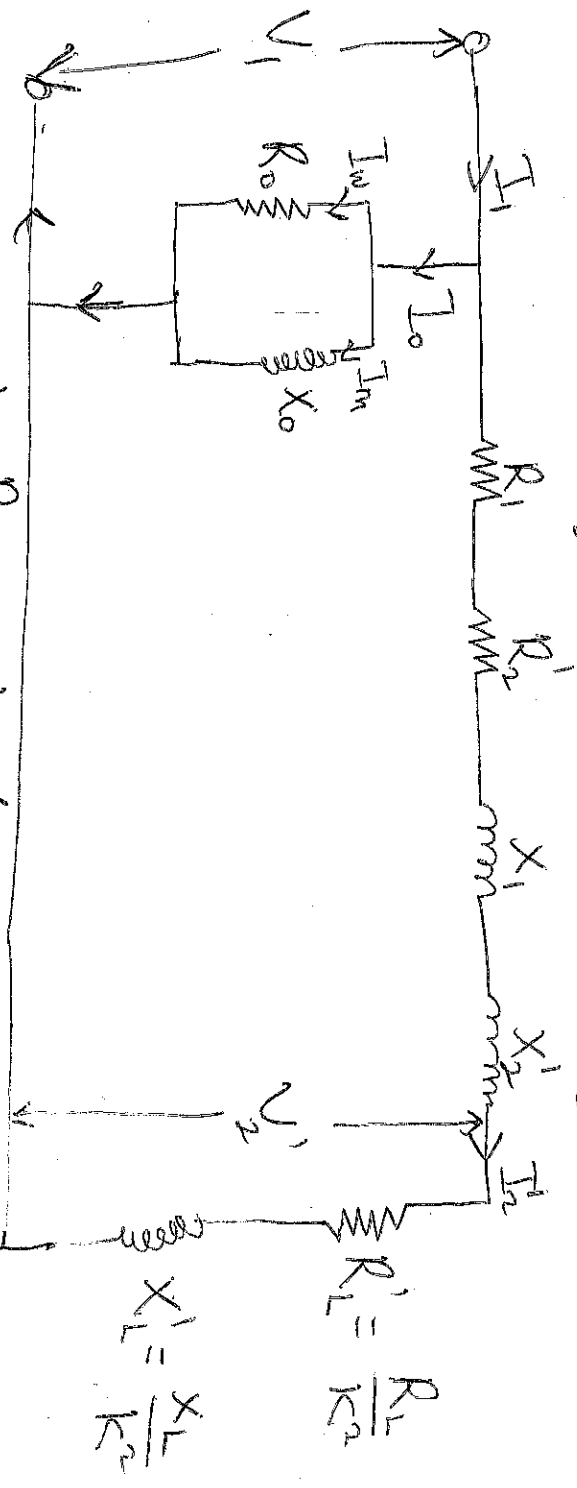


Fig 2-9 Approximate equivalent circuit of Transformer.

$$R_2' = \frac{R_2}{k^2}, \quad X_2' = \frac{X_2}{k^2}, \quad I_2' = k I_2$$

$$R_1' = \frac{R_1}{k^2}, \quad X_1' = \frac{X_1}{k^2}$$

WORKED EXAMPLE 5

- 1] The number of turns in the primary and secondary winding of a transformer are 1000 and 200 respectively. When the load current in the secondary is 100 A at 0.8 power factor lagging, the primary current is 30 A at 0.707 power factor lagging. Determine the no-load current of the transformer and its phase with respect to the voltage.

Solution: Given:

Primary turns $N_1 = 1000$

Secondary turns $N_2 = 200$

Secondary current $I_2 = 100 \text{ A}$

Power factor $\cos \phi_2 = 0.8$ or $\phi_2 = 36.9^\circ$

Primary current $I_1 = 30 \text{ A}$

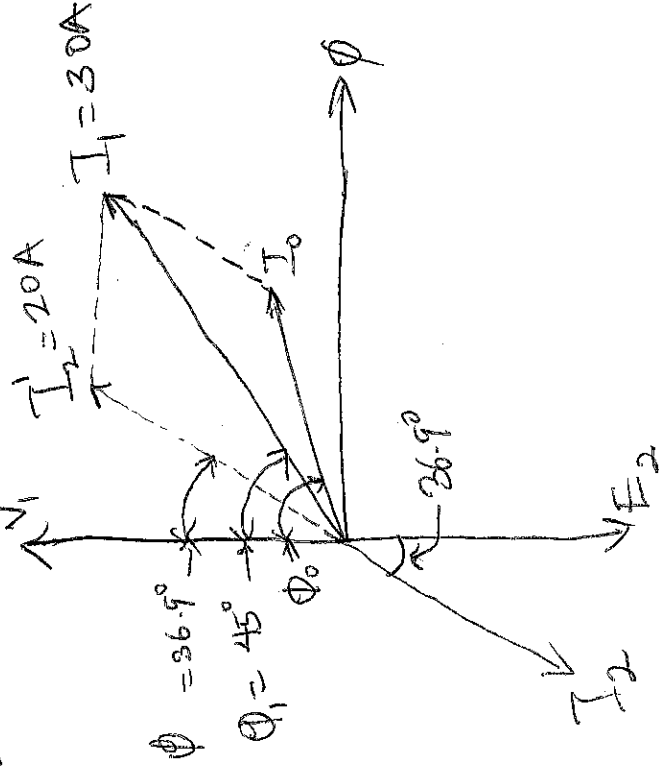
Power factor $\cos \phi_1 = 0.707$ or $\phi_1 = 45^\circ$

Find: No-load current, I_0, ϕ_0 .

Transformation ratio $k = \frac{200}{1000} = \frac{1}{5}$

Secondary current referred to primary

$$I_2' = k I_2 = \frac{1}{5} \times 100 = 20 \text{ A}$$



(7)

From the phasor diagram above,

$$I_1 = I_0 + I_2' \text{ (Vector Sum)}$$

Then Resolving currents into their x and y component

$$I_0 \cos \phi_0 + 20 \cos 36.9^\circ = 30 \cos 45^\circ$$

$$I_0 \cos \phi_0 = 30 \cos 45^\circ - 20 \cos 36.9^\circ \\ = 21.21 - 16 = 5.21 \text{ A.} \quad \text{--- (I)}$$

$$I_0 \sin \phi_0 + 20 \sin 36.9^\circ = 30 \sin 45^\circ$$

$$\therefore I_0 \sin \phi_0 = 30 \sin 45^\circ - 20 \sin 36.9^\circ \\ = 21.21 - 12 = 9.21 \text{ A} \quad \text{--- (II)}$$

From (I) & (II)

$$\frac{I_0 \sin \phi_0}{I_0 \cos \phi_0} = \tan \phi_0 = \frac{9.21}{5.21}$$

$$\text{Substituting } \phi_0 \text{ into eqn (I)} \\ \phi_0 = 60.5^\circ$$

$$I_0 \cos \phi_0 = 5.21$$

$$I_0 \cos 60.5 = 5.21$$

$$I_0 = 10.58 \text{ A.}$$

Hence, no-load current = 10.58 A

$$\phi_0 = 60.5^\circ$$

2) A 30kVA, 2000/200V, single-phase, 50Hz transformer has a primary resistance of 3.5Ω and reactance of 4.5Ω . The secondary resistance and reactance are 0.015Ω and 0.02Ω respectively. Find:

(i) Equivalent resistance, reactance and impedance referred to primary

(ii) Equivalent resistance, reactance and impedance referred to secondary

(iii) Total copper loss of the transformer.

Solution: Given:

Primary resistance, $R_1 = 3.5\Omega$

Primary reactance, $X_1 = 4.5\Omega$

Secondary resistance $R_2 = 0.015\Omega$

Secondary reactance $X_2 = 0.02\Omega$

$$\text{Transformation ratio } K = \frac{200}{2000} = \frac{1}{10} = \frac{V_2}{V_1}, \frac{I_2}{I_1}$$

(i) Equivalent resistance referred to primary

$$R_{01} = R_1 + R'_2 = R_1 + \frac{R_2}{K^2} = 3.5 + \frac{0.015}{\left(\frac{1}{10}\right)^2} = 5.0\Omega$$

(ii) Equivalent reactance referred to primary

$$X_{01} = X_1 + X'_2 = X_1 + \frac{X_2}{K^2} = 4.5 + \frac{0.02}{\left(\frac{1}{10}\right)^2} = 6.5\Omega$$

Equivalent impedance referred to primary

$$Z_{01} = \sqrt{(R_{01})^2 + (X_{01})^2} = \sqrt{5^2 + 6.5^2} = 8.2\Omega$$

(iii) Equivalent resistance referred to Secondary

$$R_{02} = R_2 + R'_1 = R_2 + K^2 R_1$$

$$= 0.015 + \left(\frac{1}{10}\right)^2 \times 3.5 = 0.05 \Omega.$$

Equivalent reactance referred to Secondary

$$X_{02} = X_2 + X'_1 = X_2 + K^2 X_1$$

$$= 0.02 + \left(\frac{1}{10}\right)^2 \times 4.5 = 0.065 \Omega.$$

Equivalent Impedance referred to Secondary

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{0.05^2 + 0.065^2} \\ = 0.082 \Omega.$$

(iv) Total Copper loss

$$\text{Secondary Current} = \frac{30 \times 1000}{200} = 150 \text{ A}$$

$$\therefore P = IV \Rightarrow 30 \text{ kVA} = 200 \text{ I} \\ \Rightarrow 30,000 = 200 \text{ I}, \Rightarrow \text{I} = \frac{30,000}{200} = 150 \text{ A}.$$

$$\text{Total Copper loss} = I_2^2 R_{02}$$

$$= 150^2 \times 0.05$$

$$= \underline{\underline{1125 \text{ W}}}.$$

Primary Current

$$P = IV$$

$$30,000 = 200 \text{ I}$$

$$\text{I} = 15 \text{ A}$$

$$\text{Total Copper loss} = I_1^2 R_{01} \\ = 15^2 \times 5 \\ = 1,125 \text{ W}.$$

TRANSFORMER TESTING

- We have the following ~~tests~~ transformer test but are concerned with the last two test.
- ① Polarity test
 - ② Voltage Ratio test.
 - ③ Sumpner's Test (or Back to Back test)
 - ④ Open-Circuit test (No-load Test)
 - ⑤ Short-Circuit test (or Impedance Test).

OPEN-CIRCUIT TEST

The purpose of this test is to determine (i) the Core Loss or excitation loss P_i (ii) No-load current and (iii) Short branch parameter R_0 & X_0 of the equivalent

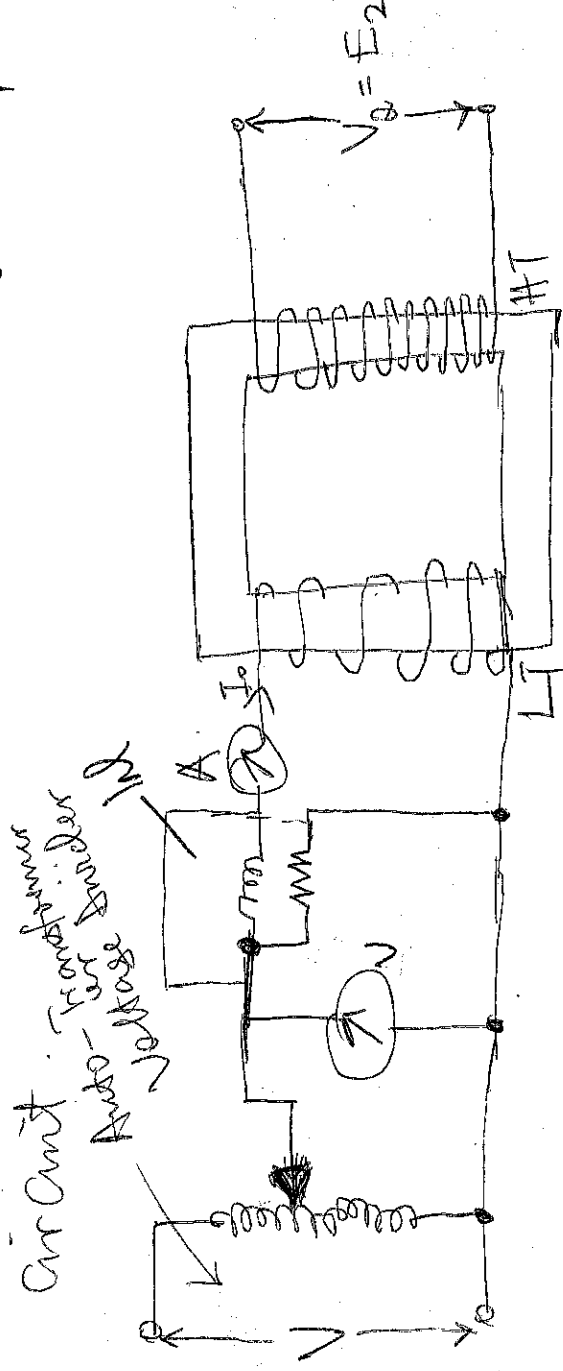


Fig 2-9 Open-Circuit Test Circuit Diagram

Fig 2-9 above depicts the circuit diagram of open circuit test. In this test the secondary winding (HV) is kept open circuited and the rated voltage at the rated frequency is applied to the primary winding. The core loss will be the same whether the measurement is taken from the primary or the secondary winding, but the voltage would be large and the current small if the measurement is taken from the secondary side.

⑦

In auto-transformer or Voltage divider is used to varying the voltage applied to the primary winding. Ammeter A and Voltmeter V are connected to measure no-load current I_0 and input power P_0 . Voltmeter V is connected to measure the applied voltage.

The open circuit test gives enough data to compute the equivalent circuit constants R_0, X_0 , no-load power factor $\cos \phi$, no-load current I_0 and no-load power loss (Iron loss) of a transformer.

Iron loss, P_i = Input power on no load = P_0 (Watts) = $I_0 V_1$
 No-load current = I_0 (amperes) $\therefore I_e = \frac{P_0}{V_1}$

Applied voltage to primary = V_1 (Volts)
 Angle of $P_0, \phi_0 = \cos^{-1} \frac{P_0}{V_1 I_0}$ No load power factor $\cos \phi_0 = P_0 / V_1 I_0$

No load current energy component, $I_e = I_0 \cos \phi_0 = \frac{P_0}{V_1}$

No load current magnetizing component, $I_m = \sqrt{I_0^2 - I_e^2}$ or $I_0 \sin \phi$
 Equivalent circuit parameter, $R_0 = \frac{V_1}{I_e} = \frac{V_1^2}{P_0}$

Equivalent circuit parameter, $X_0 = \frac{V_1}{I_m} = \frac{V_1}{\sqrt{I_0^2 - I_e^2}}$

SHORT-CIRCUIT TEST [OR IMPEDANCE TEST]

The purpose of this test is to (i) determine full load copper loss (W) Equivalent resistance (ii) Equivalent reactance referred to metering side.

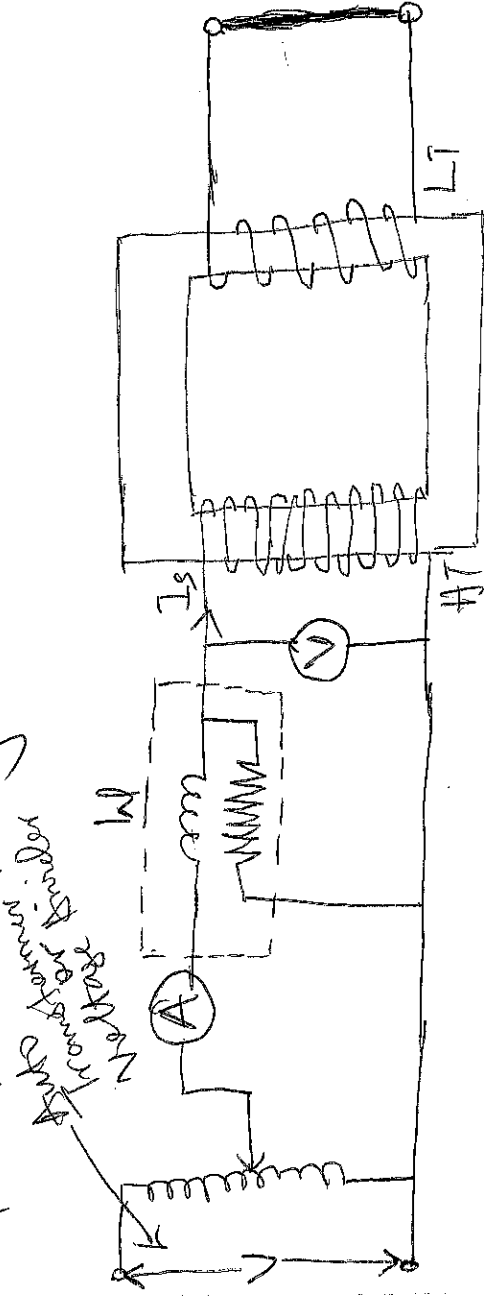


Fig 2-10 Short-Circuit Test Circuit Diagram.

Fig 2-10 depicts the short-circuit test circuit diagram of a transformer. In this test, the terminal of a secondary winding, (L-side) are short-circuited by a thick wire or strip or through an ammeter. Variable low voltage is applied to the primary through an auto transformer or voltage divider. The power input (reading of wattmeter) gives total copper loss at rated load, output being nil. Let the reading of voltmeter, ammeter and wattmeter be V_s , I_s and W_s respectively.

$$\text{Full-load copper loss } P_c = I_s^2 R_{eq} = W_s \approx P$$

$$\text{Equivalent resistance, } R_{eq} = \frac{W_s}{I_s^2} = \frac{P_s}{I_s^2}$$

$$\text{Equivalent impedance, } Z_{eq} = \frac{V_s}{I_s}$$

$$\text{Equivalent reactance, } X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

Model Example

1] The primary of a transformer is rated at 10A and 11,000V. On open-circuit the readings are $V_1 = 1,000V$, $V_2 = 500V$, $I = 0.42A$ and $P_o = 100W$. On short-circuit the readings are $I_1 = 10A$, $V_1 = 126V$ and $P_{sc} = 400W$. Draw an equivalent circuit for the transformer and determine the parameters. Predict the output voltage across a load impedance $Z_L = (19 + j12)\Omega$.

SolnNo-load current, $I_o = 0.42A$ No-load input power, $P_o = 100W$ Primary Voltage, $V_1 = 1000V$ No-load power factor, $\cos \phi_o = \frac{P_o}{V_1 I_o} = \frac{100}{1000 \times 0.42} = 0.2381$ Energy component of no-load current, $I_e = I_o \cos \phi_o$
 $= 0.42 \times 0.2381$
 $= 0.1A$

Magnetising component of no-load current

$$I_m = \sqrt{I_o^2 - I_e^2} = \sqrt{(0.42)^2 - (0.1)^2}$$

$$= 0.408A$$

No-load resistance, $R_o = \frac{V_1}{I_e} = \frac{1000}{0.1} = 10k\Omega$ No-load reactance $X_o = \frac{V_1}{I_m} = \frac{1000}{0.408} = 2.45k\Omega$ Short-circuit current, $I_s = 10A$ Short-circuit Voltage, $V_s = 126V$ Power input on short-circuit, $P_s = 400W$

Equivalent impedance referred to primary

$$Z_{e1} = \frac{V_s}{I_s} = \frac{126}{10} = 12.6\Omega$$

Equivalent resistance referred to primary

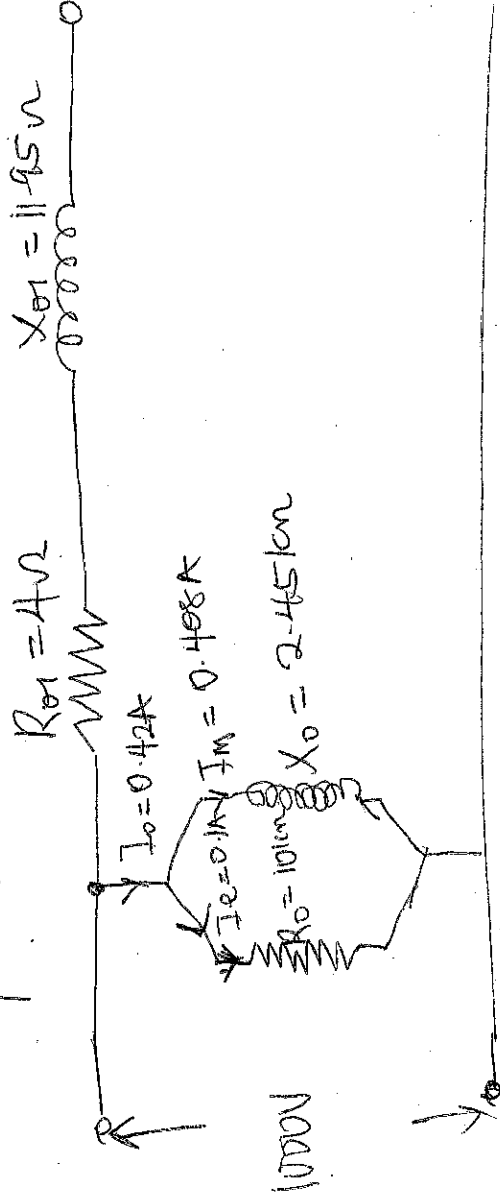
$$R_{01} = P_s / I_s^2 = \frac{400}{(10)^2} = 4 \Omega$$

Equivalent reactance referred to primary

$$X_{01} = \sqrt{(Z_{01})^2 - (R_{01})^2}$$

$$= \sqrt{(12.6)^2 - (4)^2} = 11.95 \Omega$$

The equivalent circuit is shown below



Since, Open circuit voltage $V_2 = 500V$ as given

$$\text{Transformation ratio } k = \frac{V_2}{V_1} = \frac{500}{1000} = 0.5$$

Equivalent resistance referred to secondary

$$R_{02} = k^2 R_{01}$$

$$= 0.5^2 \times 4 = 1 \Omega$$

Equivalent reactance referred to secondary, $X_{02} = k^2 X_{01}$

$$= (0.5)^2 \times 11.95 = 2.9875 \Omega$$

Equivalent impedance referred to secondary

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} \Rightarrow 1 + j2.9875 = Z_{02}$$

Load impedance $Z_L = (19 + j12) \Omega$

$$\text{Load voltage} = \frac{V_2 \times Z_L}{Z_{02} + Z_L} = \frac{500 \times (19 + j12)}{(1 + j2.9875) + (19 + j12)}$$

$$= \frac{500 \times 22.47 \angle 32.3^\circ}{25 \angle 36.85^\circ} = 449.4 \angle -4.55^\circ$$

$$\begin{aligned} 19 + j12 &= \sqrt{19^2 + 12^2} \angle \tan^{-1}(12/19) \\ &= 22.47 \angle 32.3^\circ \end{aligned}$$

EFFICIENCY AND VOLTAGE REGULATION OF TRANSFORMER (11)

VOLTAGE REGULATION.

$$\% \text{ Regulation} = \frac{\text{Terminal Voltage on no load} - \text{Terminal Voltage on load}}{\text{Terminal Voltage on no load}} \times 100$$

$$= \frac{\text{Voltage drop in transformer at load}}{\text{No-load rated voltage (Secondary)}} \times 100$$

$$= \frac{I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi}{\text{No-load rated voltage (Secondary)}} \times 100$$

When Power factor is leading, the Percentage regulation is given by

$$= \frac{I_2 R_{02} \cos \phi - I_2 X_{02} \sin \phi}{\text{No-load rated Secondary Voltage}} \times 100$$

Condition for Zero Regulation

$$\text{Regulation} = \frac{I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi}{E_2}$$

Regulation will be zero if the numerator will be equal to zero or $I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi = 0$

$$\text{or } \tan \phi = -\frac{R_{02}}{X_{02}}$$

Since $\tan \phi$ is $\frac{\sin \phi}{\cos \phi}$

Condition for maximum Regulation

Regulation will be maximum if $\frac{d}{d\phi} (\text{regulation}) = 0$

$$\text{or } \frac{d}{d\phi} \frac{I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi}{E_2} = 0$$

$$\text{or } -\frac{I_2 R_{02}}{E_2} \sin \phi + \frac{I_2 X_{02}}{E_2} \cos \phi = 0$$

$$\text{or } \tan \phi = \frac{X_{02}}{R_{02}}$$

TRANSFORMER EFFICIENCY

The efficiency of a transformer at a particular load and power factor is defined as the ratio of power output to power input.

$$\therefore \text{Efficiency} = \frac{\text{Output}}{\text{Input}}$$

$$= \frac{\text{Output}}{\text{Output} + \text{losses}} = \frac{\text{Output}}{\text{Output} + \text{Cu loss} + \text{Iron loss}}$$

$$\text{Efficiency} = \frac{\text{Input} - \text{losses}}{\text{Input}} = 1 - \frac{\text{losses}}{\text{Input}}$$

* Efficiency can be calculated by determining core losses from Open-Circuit test and Copper losses from Short-Circuit test.

Condition for Maximum Efficiency

Iron losses P_i = hysteresis loss + eddy current loss
 $= P_h + P_e$

$$\text{Copper loss } P_c = I_1^2 R_1 \text{ or } I_2^2 R_2$$

Considering Primary Side

$$\text{Input to primary} = V_1 I_1 \cos \phi_1$$

$$\text{Efficiency } \eta = \frac{V_1 I_1 \cos \phi_1 - \text{losses}}{V_1 I_1 \cos \phi_1}$$

$$= \frac{V_1 I_1 \cos \phi_1}{V_1 I_1 \cos \phi_1 - I_1^2 R_1 - P_i}$$

$$= \frac{V_1 I_1 \cos \phi_1}{V_1 I_1 \cos \phi_1 - I_1^2 R_1 - P_i}$$

$$= 1 - \frac{I_1 R_1}{V_1 \cos \phi_1} - \frac{P_i}{V_1 I_1 \cos \phi_1}$$

* How would you determine the efficiency of a transformer?
* Show that the constant current of transformer corresponding to maximum efficiency is $I_2 = \sqrt{P_i / R_{02}}$.

(12)

$$\begin{bmatrix} nI_1^{n-1} \\ -1I_1^{n-1} \\ -1I_2^{n-1} \\ -1I_2^{n-1} \end{bmatrix}$$

Differentiate both sides w.r.t. I_1 , we get

$$\frac{d^2}{dI_1} = 0 - \frac{R_{e1}}{V_1 \cos \phi_1} + \frac{P_i}{V_1 I_1^2 \cos \phi_1}$$

For 2 to be maximum

$$\frac{d^2}{dI_1} = 0 \quad \text{Hence above equation reduces to}$$

$$\frac{R_{e1}}{V_1 \cos \phi_1} = \frac{P_i}{V_1 I_1^2 \cos \phi_1}$$

$$P_i = I_1^2 R_{e1} \text{ or } I_2^2 R_{o2}$$

or Copper losses = Iron losses

The Output Current Corresponding to maximum efficiency is

$$I_2 = \sqrt{\frac{P_i}{R_{o2}}}$$

Worked Example.

□ The following readings were obtained from

O.C and S.C test on 8 KVA 440/120V 50Hz transformer

O.C test (l.v. side)

S.C test (h.v. side)

120V, 4A, 75W
9.5V, 20A, 110W

Calculate: (i) The equivalent circuit (approximate) constants.

(ii) Voltage regulation and efficiency for 0.8 lagging

power factor load

* (iii) The efficiency at half full-load and 0.8 power factor load.

Solution O.C test

Transformation ratio $K = \frac{120}{400} = \frac{3}{10}$
No-load current $I_0 = 4 \times \frac{3}{10} = 1.2 \text{ A}$ $\therefore I_2 = K I_1$

$$P_0 = V_1 I_0 \cos \phi_0$$

$$75 = 400 \times 1.2 \cos \phi_0$$

$$\therefore \cos \phi_0 = \frac{75}{400 \times 1.2} = 0.156$$

$$\sin \phi_0 = 0.987$$

Watt full component of no-load current, I_0

$$I_e = I_w = I_0 \cos \phi_0 = 1.2 \times 0.156 = 0.187 \text{ A}$$

Magnetising component of no-load current, I_0

$$I_m = I_0 \sin \phi_0 = 1.2 \times 0.987 = 1.184 \text{ A}$$

or $I_m = \sqrt{I_0^2 - I_w^2} = 1.185 \text{ A}$

$$\text{Core loss } P = I_w V_1 = I_w^2 R_s$$

Resistance representing the loss

$$R_o = \frac{V_1}{I_w} = \frac{400}{0.187} = 2139 \Omega$$

Magnetising reactance $X_0 = \frac{V_1}{I_m} = \frac{400}{1.184} = 337.8 \Omega$

S.C. test

During S.C test the instruments have been placed in primary
Here

$$V_{sc} = 9.5 \text{ V}, I_{sc} = 20 \text{ A}, P_{sc} = 110 \text{ W}$$

$$Z_{01} = \frac{V_{sc}}{I_{sc}} = \frac{9.5}{20} = 0.475 \Omega$$

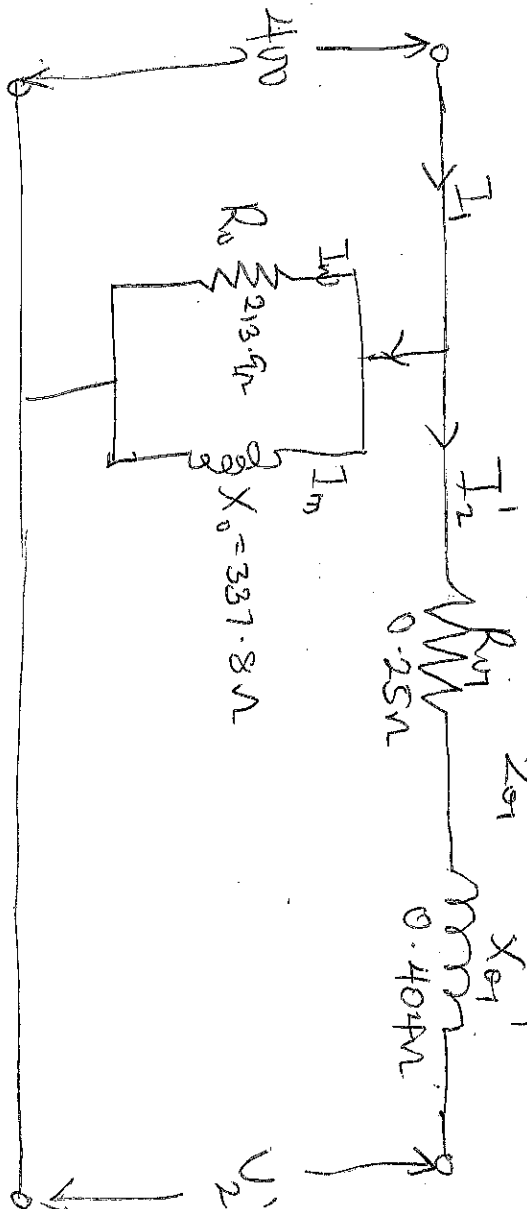
$$P_{sc} = I_{sc}^2 R_{01} \Rightarrow R_{01} = \frac{P_{sc}}{I_{sc}^2} = \frac{110}{(20)^2} = 0.25 \Omega$$

$$Z_{01} = \sqrt{(R_{01})^2 + (X_{01})^2}$$

$$\therefore X_{01} = \sqrt{(Z_{01})^2 - (R_{01})^2} = \sqrt{0.475^2 - 0.25^2}$$
$$= 0.404 \Omega$$

(13)

The equivalent circuit is shown below



① Total approximate voltage drop as referred to primary is
 $= I_1 (R_1 \cos \phi + X_1 \sin \phi)$

Full-load current $I_1 = \frac{8 \text{ kVA}}{400} = \frac{8 \times 1000}{400}$
 $= 20 \text{ A}$

$\therefore$ Voltage drop $= 20 (0.25 \times 0.8 + 0.404 \times 0.6) = 8.85 \text{ V}$
 $\left[\begin{array}{l} \cos \phi = 0.8 \quad \phi = \cos^{-1} 0.8 = 0.6435 \\ \text{then } \sin \phi = \sin(0.6435) = 0.5999 \approx 0.6 \end{array} \right]$

$\therefore$ Voltage Regulation $= \frac{8.85}{400} \times 100 = 2.21\%$

Iron loss $= P_i = 75 \text{ W}$

Copper loss $P_c = 110 \text{ W}$

Full-load losses $= P_i + P_c = 75 + 110$
 $= 185 \text{ W}$

$\therefore$ Full-load Output $= 8 \times 1000 \times 0.8 = 6400 \text{ W}$

Full-load efficiency

$\eta = \frac{\text{Output}}{\text{Output} + \text{losses}} = \frac{6400}{6400 + 185}$
 $= 0.972 \text{ or } 97.2\%$

THREE PHASE TRANSFORMER

The voltage transformation ~~is~~ in a 3-phase network can be achieved either by means of a bank of three single phase transformer or a single 3-phase transformer.

Advantages of three-phase over single-phase transformer

- 1) The same iron is used for interlinking the magnetic circuit. ~~thereby reduces the~~
- 2) Proper flux densities are maintained because the 3-phase currents are displaced 120 electrical degrees with respect to each other.
- 3) Less transformer oil is used since the entire three phase assembly is reduced in size and is placed in single tank.
- 4) Fewer transformer bushing are required.
- 5) The complete transformer occupies less floor space than a bank of a three single phase transformers.
- 6) A 3-phase transformer is cheaper than a bank of three single phase transformers.
- 7) A 3-phase transformer is slightly more efficient than a bank of a three single phase transformers because it has a shorter magnetic path and core volume and hence the ~~core~~ core loss is smaller.
- 8) The installation and operational cost of a 3-phase transformer is smaller because there is only one unit to install and operates.

Conclusion

Why is 3- ϕ transformer more efficient than a bank of 3 single phase transformer.

Disadvantage of 3-Phase Transformer Over Single phase (14)

- 1) It is larger and heavier than any other single phase transformer and is more difficult to handle, ship and set in place for operation.
- 2) 3-phase transformers are much more difficult and costly to repair than single phase unit.

A 3- ϕ transformer can be constructed either by connecting together three single phase transformers thereby forming a so called three phase transformer bank or by using one pre-assembled and balanced three phase transformer which consists of three pairs of single phase windings mounted onto one single laminated core.

If we take three single-phase transformers and connect their primary winding to each other in a fixed configuration, we windings to each other in a fixed configuration, we can use the transformers on a three-phase supply.

3- ϕ Transformer Connections

The primary and secondary windings of a transformer can be connected in different configuration as shown in the diagram below to meet practically any requirement.

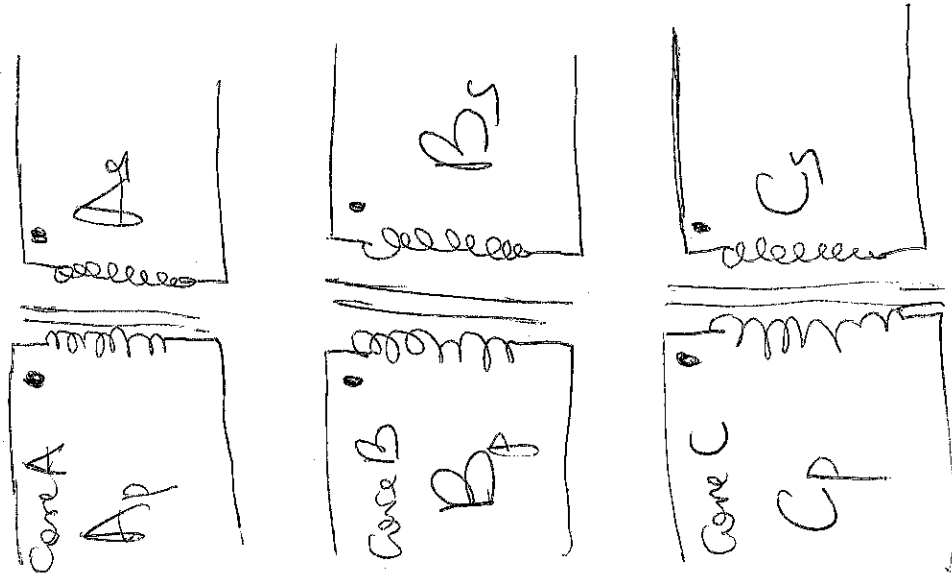
In the case of 3- ϕ transformer windings, three forms of connections are possible viz

- (i) Star (ii) Delta (iii) Interconnected Star or

29-209.

The combination of the three windings may be with the primary delta-connected and the secondary Star-connected, or Star-delta, Star-Star or delta-delta depending on the transformer use.

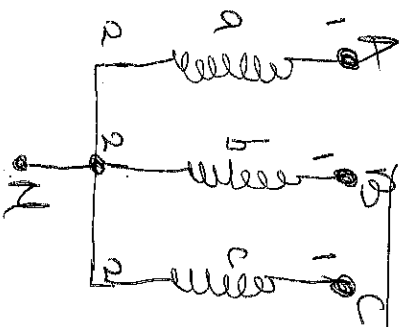
3 Single Phase Transformer



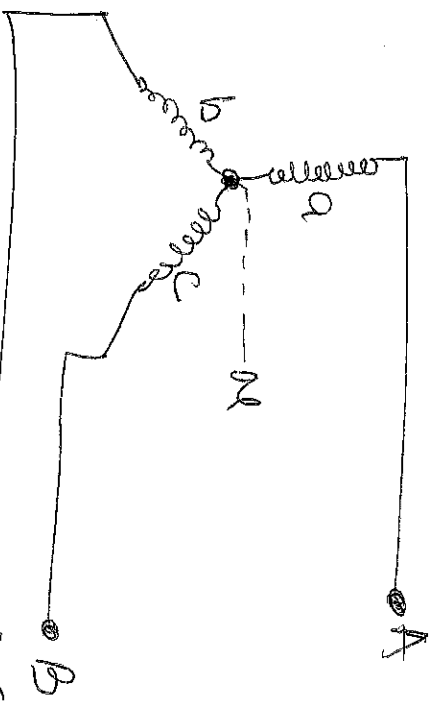
Primary Configurations	Secondary Configurations
Delta (Mesh) Δ	Delta (Mesh) Δ
Delta (Mesh) Δ	Star (Wye) Y
Star (Wye) Y	Delta (Mesh) Δ
Star (Wye) Y	Star (Wye) Y
Interconnected Star	Delta Δ
Interconnected Star	Star Y

The standard method for marking three phase transformer windings is to label the three phase primary windings with Capital (upper case) letters A, B and C, used to represent the three individual phases of Red, Yellow and Blue. The Secondary windings are labelled with small (lower case) letter a, b, and c. Each winding has two ends normally labelled 1 and 2 so that for example, the Second winding of the primary has ends which will be labelled B1 and B2 while the third winding of the Secondary will be labelled C1 and C2 as shown in the diagram below.

Star-Connection

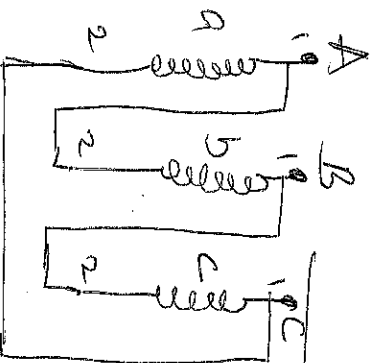


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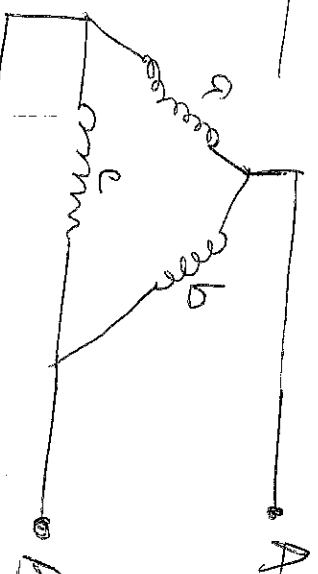


(15)

Delta-Connection



$\equiv$



Symbols are generally used on a three phase xformer to indicate the type or types of connections used with upper case Y for Star Connected, Δ for delta Connected and Z for Interconnected stars Primary winding, with lower case y, d and z for their respective secondaries. Then Star-Star would be labelled Yy, Delta-Delta $\Delta\Delta$ and Inter Connected Star to Inter Connected Star Zz Transformer winding Identification

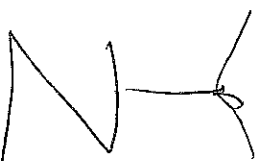
Connection

Delta

Star

Interconnected

Primary Winding



Secondary Winding

d

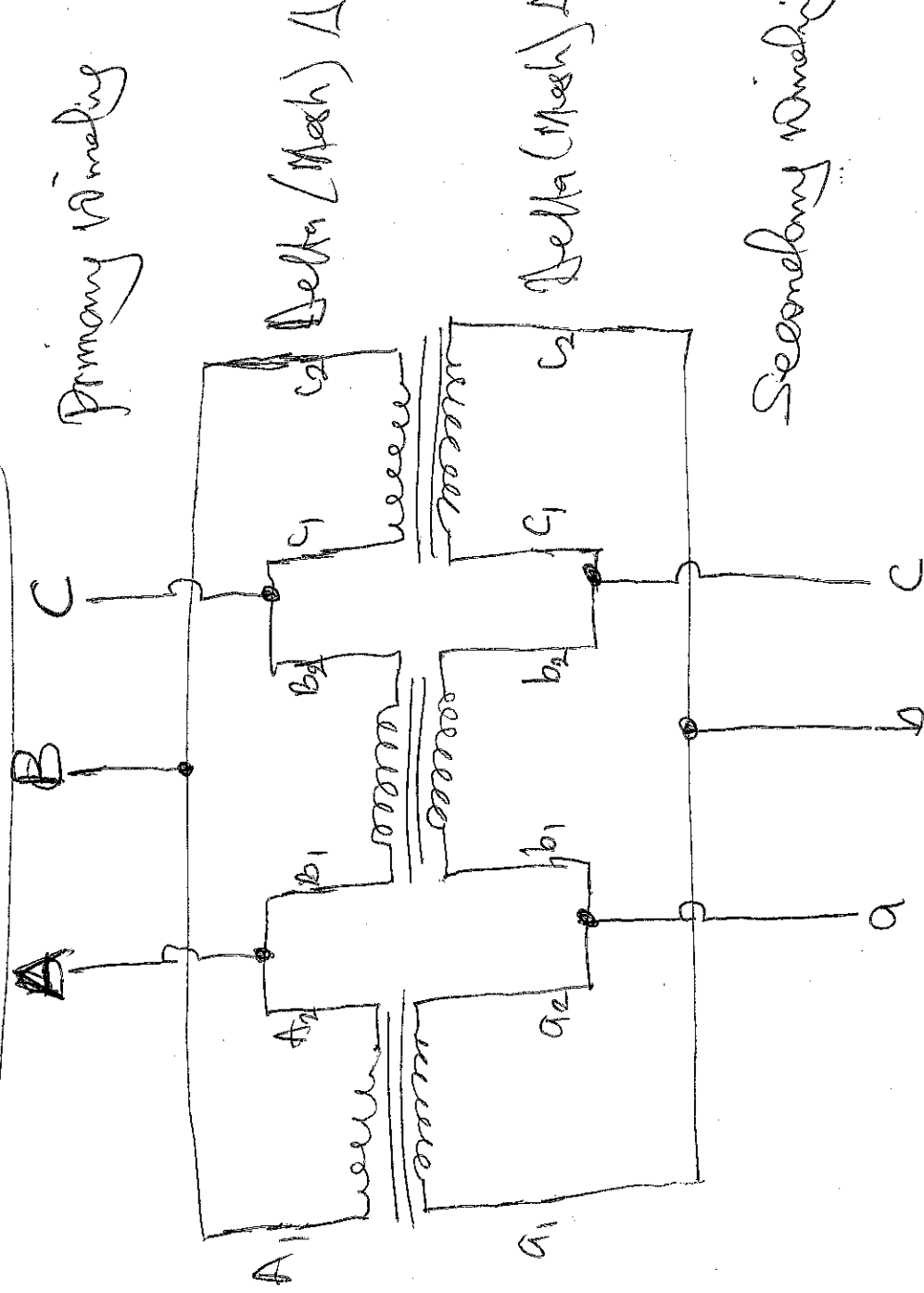
y

z

There are four different ways in which three single phase transformers may be connected between their primary and secondary to form three phase circuits. These four standard configurations are given as

- ① Delta-Delta $\Delta\Delta$
- ② Star-Star Y_Y
- ③ Star-Delta $Y\Delta$
- ④ Delta-Star ΔY

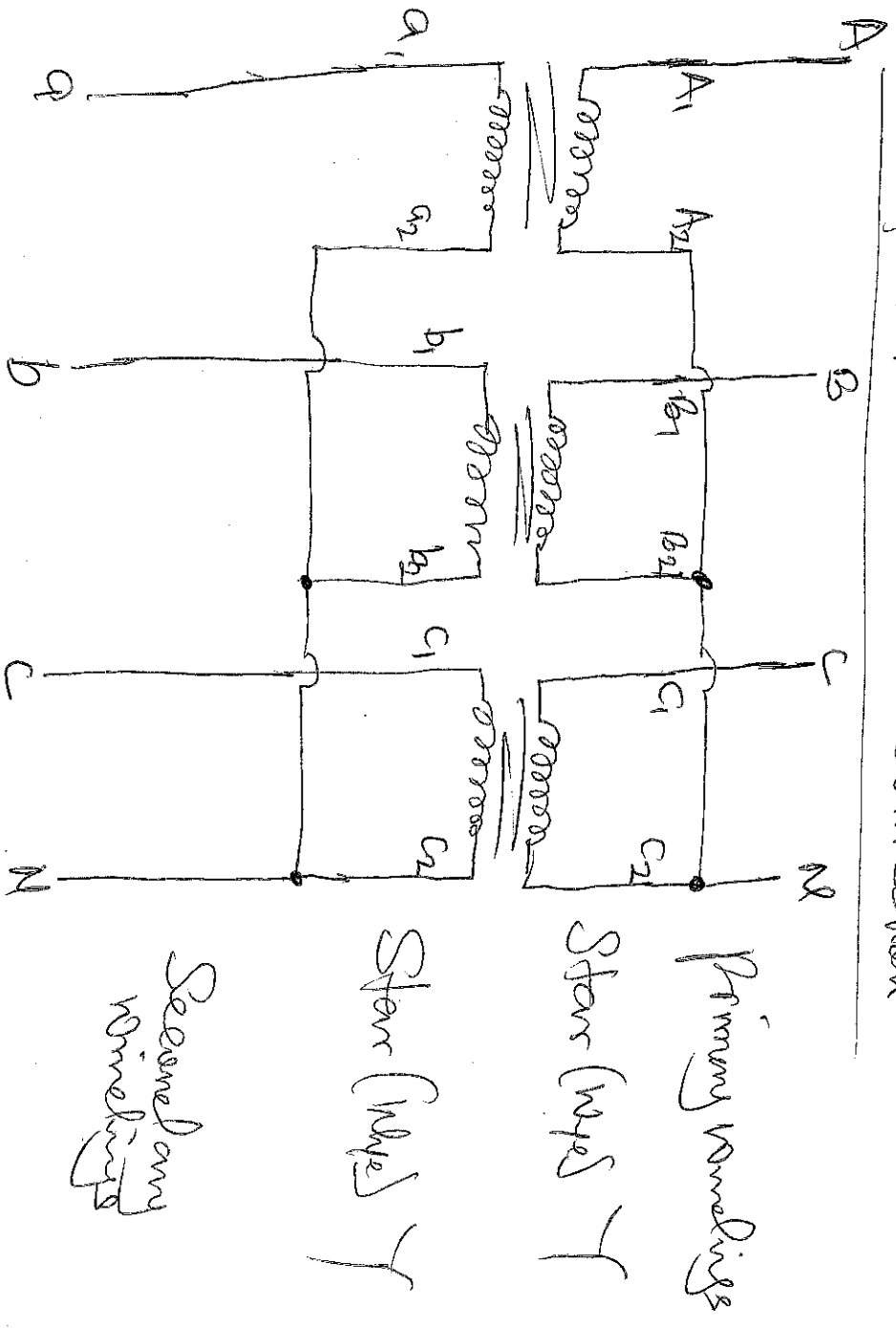
Transformer Delta and Delta Connection



In delta connection ($\Delta\Delta$) group of former, the line voltage, $V_L = \text{Supply Voltage } V_S$ i.e. $V_L = V_S$
 Current in each phase = $\frac{1}{\sqrt{3}} \times I_L$ where I_L is the line current.

Transformer Star-Star Connection

(16)



The voltage between any line of the three-phase transformer is called the line voltage, V_L while the voltage between any line and the neutral point of a star connected transformer is called the phase voltage, V_p .

$$\therefore V_p = \frac{1}{\sqrt{3}} \times V_L$$

The secondary current in each phase of a star connected group of transformers is the same as that for the line current of the supply, then $I_L = I_s$

Relationship between line and phase voltage and current in a 3- ϕ system can be summarised below

Connection	Phase Voltage	Line Voltage	Phase Current	Line Current
Star	$V_p = \frac{1}{\sqrt{3}} \times V_L$	$V_L = \sqrt{3} V_p$	$I_p = I_L$	$I_L = I_p$
Delta	$V_p = V_L$	$V_L = V_p$	$I_p = \frac{1}{\sqrt{3}} \times I_L$	$I_L = \sqrt{3} \times I_p$

When the primary and secondary have different types of winding connection, star or delta, the overall turns ratio of transformers becomes more complicated. For the four basic configurations of a 3- ϕ transformer, we can list the former secondary voltages and currents with respect to the primary line voltage, V_L and its primary line current I_L as shown below

Primary to Secondary	Line Voltage Primary or Secondary	Line Current Primary or Secondary
Delta-Delta	$V_L = n V_L$	$I_L = \frac{I_L}{n}$
Delta-Star	$V_L = \sqrt{3} \cdot n \cdot V_L$	$I_L = \frac{I_L}{\sqrt{3} \cdot n}$
Star-Delta	$V_L = \frac{1}{\sqrt{3}} \times n V_L$	$I_L = \sqrt{3} \cdot \frac{I_L}{n}$
Star-Star	$V_L = n V_L$	$I_L = \frac{I_L}{n}$

Worked Example

(17a)

1] The primary winding of a delta-star (Dy) Connected 50VA transformer is supplied with a 100V, 50Hz three phase supply. If the transformer has 500 turns on the primary and 100 turns on the secondary winding. Calculate the secondary side voltages and current.

Soln

Given: Transformer rating = 50VA

Supply Voltage, $V_s = 100V$

Primary turns

= 500

Secondary turns

= 100

$$\text{Transformation ratio, } n = \frac{N_s}{N_p} = \frac{100}{500} = 0.2$$

$$V_L(\text{sec}) = \sqrt{3} \times n \times V_L(\text{pri}) \\ = \sqrt{3} \times 0.2 \times 100 = 34.64V$$

$$V_p(\text{sec}) = \frac{V_L(\text{sec})}{\sqrt{3}} = \frac{34.64}{\sqrt{3}} \approx 20\text{Volts}$$

$$I_L(\text{pri}) = \frac{VA}{\sqrt{3} V_L(\text{pri})} = \frac{50}{\sqrt{3} \times 100} = 0.289A$$

$$I_{\text{sec}} = \frac{I_L(\text{pri})}{\sqrt{3} \times n} = \frac{0.289}{\sqrt{3} \times 0.2} = 0.834A$$

worked Example 2

A 3-phase Step down transformer takes 15A when connected to 4400V mains. The turn ratio per phase is 10. Neglecting losses find the Secondary line Voltage, line Current and output if the transformer windings are connected in (i) Star / delta (ii) delta / star.

Solution

$$\text{Turn ratio/phase, } \frac{N_1}{N_2} = 10, K = \frac{N_2}{N_1} = \frac{1}{10}$$

(i) Star / delta

Primary is connected in Star and Secondary in delta.

$$\begin{aligned}\text{Primary phase Voltage} &= \frac{1}{\sqrt{3}} \times \text{line Voltage} \\ &= \frac{1}{\sqrt{3}} \times 4400 = 2540\text{V}\end{aligned}$$

$$\begin{aligned}\text{Secondary phase Voltage} &= K \times 2540 \\ &= \frac{1}{10} \times 2540 \\ &= 254\text{V}\end{aligned}$$

$$\text{Secondary line Voltage} = \text{Phase Voltage} = 254\text{V}$$

$$\begin{aligned}\text{Primary phase Current} &= \text{line Current} = 15\text{A} \\ \text{Secondary phase Current} &= 15 \times \frac{1}{K} = 15 \times 10 = 150\text{A} \\ \text{Secondary line Current} &= \sqrt{3} \times \text{phase Current} \\ &= \sqrt{3} \times 150 \\ &= 259.8\text{A}.\end{aligned}$$

$$\begin{aligned}\text{Output in kVA} &= \sqrt{3} \times 254 \times 259.8 \times 10^{-3} \\ &= 114.29\text{kVA}.\end{aligned}$$

⑪ Delta / Star

Primary is connected in delta and Secondary in Star

Primary Phase Voltage = Line Voltage = 4400 V

Secondary Phase Voltage = $K \times 4400$

$$= \frac{1}{10} \times 4400 = 440 \text{ V}$$

Secondary line Voltage = $\sqrt{3} \times \text{Phase Voltage}$

$$= \sqrt{3} \times 440 = 762 \text{ V}$$

Primary Phase Current = $\frac{1}{\sqrt{3}} \times \text{Line Current}$

$$= \frac{1}{\sqrt{3}} \times 15 = 8.66 \text{ A}$$

Secondary Phase Current = $\frac{1}{K} \times 8.66$

$$= \frac{1}{10} \times 8.66$$

$$= 10 \times 8.66$$

$$= 86.6 \text{ A}$$

Secondary line Current = Phase Current
= 86.6 A

$$\begin{aligned} \text{Output in kVA} &= \sqrt{3} \times 762 \times 86.6 \times 10^{-3} \\ &= \underline{\underline{114.29 \text{ kVA}}} \end{aligned}$$

Assignment

Solve the above problem when the transformer is connected in Δ/Δ Star / Star. ⑫ Delta / Delta.

TERTIARY WINDINGS

When a transformer is built with a third winding called tertiary in addition to the normal winding (primary and secondary) such a transformer is called the triple-wound (or 3-winding transformer).

Purpose of Tertiary Winding

- 1) To supply the substation auxiliaries ^{at a} different from those of primary and secondary winding.
- 2) To supply phase compensating devices, such as Condenser operated at a voltage which is different from that of primary and secondary winding.
- 3) To interconnect three supply systems operating at different voltage.
- 4) To load large split-winding generator.
- 5) To measure voltage of an HV testing transformer.

The tertiary winding is called auxiliary winding when it is employed for supplying an additional small load at a different voltage.

It is called stabilizing winding when it is employed to limit the short-circuit current.

Parallel Operation of Transformers

This is an arrangement in which the primaries of two or more transformers energized from the same source (i.e. the same generator or the same output winding of a bigger transformer or voltage regulator) and the secondaries are connected to supply the same load.

Important of Parallel Operation

- 1] If the amount of power to be transformed is greater than that which can be handled by one transformer, it is necessary to employ one or more transformers in parallel.
- 2] To meet the additional demand on power system, it becomes necessary either to replace the existing unit by unit of greater capacity or to install additional unit to carry a portion of the load. The cost of making such changes will in many cases be greater than if the original transformer is connected in parallel with another to carry a portion of the load.

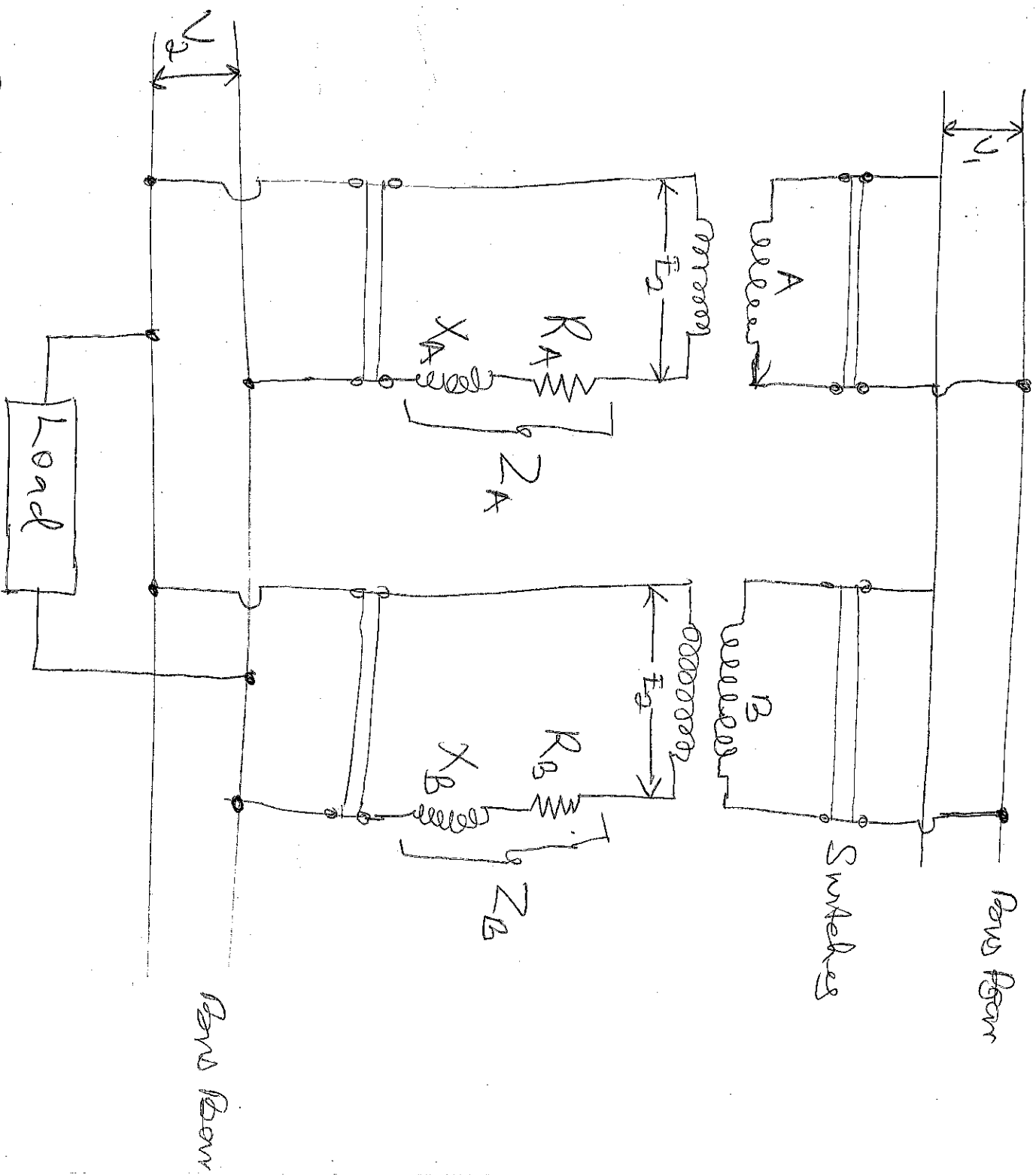
Disadvantage

The main drawback of operating transformer in parallel is that there will be an increase in short-circuit current sometimes to a level beyond the interrupting capacity of the existing over-current device.

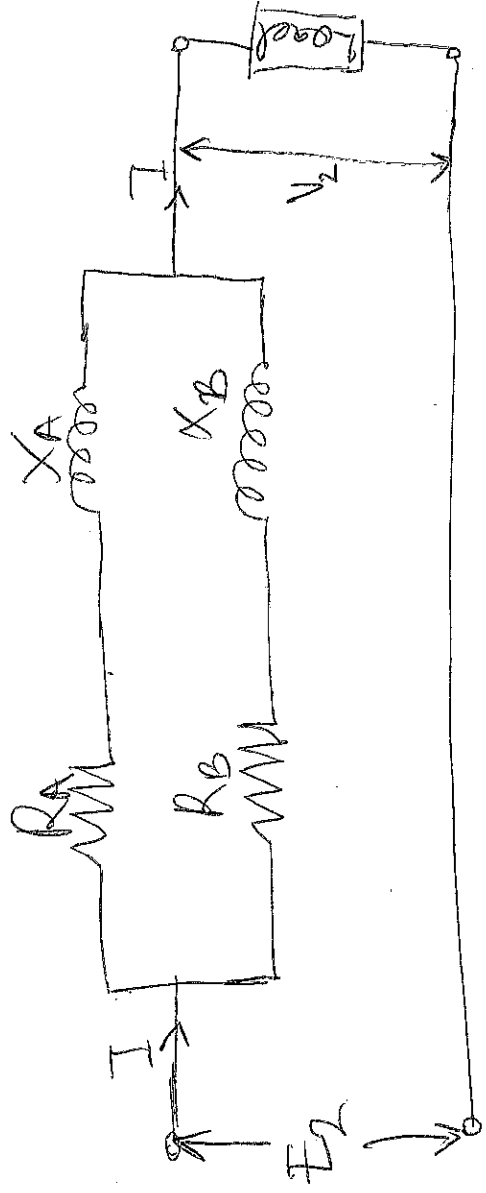
Conditions for Parallel Operation of Transformer

- i] The voltage ratio must refer to the terminal voltage of p.r. & see.
- ii] The phase displacement b/w p.r. & see voltages must be the same for all transformers which are to be paralleled. (i) The equivalent impedances should be inversely proportional to the phase sequence must be the same respectively ~~not~~ rating.
- iii] The ratio of the equivalent resistance to equivalent reactance of all transformers must be the same.

Load Sharing Between Two Transformers (17) Operating in Parallel



Circuit Diagram of Transformer Connected in Parallel



Approximate Equivalent Circuit of Two Transformer Operating in Parallel

from equivalent circuit diagram above, two voltage equations may be written:

$$E_2 = V_2 + I_A Z_A \quad \text{--- (1)}$$

$$\text{and } E_2 = V_2 + I_B Z_B \quad \text{--- (2)}$$

$$\text{Or } I_A Z_A = I_B Z_B \quad \text{--- (3)}$$

$$\text{Or } \frac{I_A}{I_B} = \frac{Z_B}{Z_A} = \frac{R_B + jX_B}{R_A + jX_A} \quad \text{--- (4)}$$

where R_A , X_A and Z_A are total resistance, reactance and impedance of transformer A and R_B , X_B and Z_B for transformer B.

$$\text{Also, } I_A Z_A = I_B Z_B = I Z_{AB} \quad \text{--- (5)}$$

where I is the total current and Z_{AB} is the combined impedance of Z_A and Z_B in parallel.

$$\text{i.e. } Z_{AB} = \frac{Z_A Z_B}{Z_A + Z_B} \quad \text{--- (6)}$$

from eqn (5)

$$I_A = \frac{I Z_{AB}}{Z_A} \quad \text{--- (7)}$$

$$= I * \frac{Z_A Z_B}{Z_A + Z_B} * \frac{1}{Z_A} \quad \text{--- (8)}$$

$$I_A = I \frac{Z_B}{Z_A + Z_B} \quad \text{--- (9)}$$

Similarly

$$I_B = I \frac{Z_A}{Z_B + Z_A} \quad \text{--- (10)}$$

Multiply both sides of equation (9) and (10) by V_2

$$V_2 I_A = V_2 I \frac{Z_B}{Z_A + Z_B} \quad \text{Volt-ampere. } P$$

or Load shared by transformer A

$$S_A = V_2 I_A \times 10^{-3} \text{ kVA} \quad \text{loss}$$

$$= V_2 I \frac{Z_B}{Z_A + Z_B} \times 10^{-3} \text{ kVA}$$

$$= S \frac{Z_B}{Z_A + Z_B} \quad \text{--- (11)}$$

and Load shared by transformer B:

$$S_B = V_2 I \frac{Z_A}{Z_B + Z_A} \times 10^{-3} \text{ kVA}$$

$$= S \frac{Z_A}{Z_B + Z_A} \quad \text{[where } S = V_2 I \times 10^{-3}]$$

TRANSFORMER NOISE

Under no load conditions, the "hum" caused by energized power transformer originates in the cores, where the lamination tend to vibrate by magnetic forces. The noise is transmitted through the oil to the tank side and then to the surrounding. Factors responsible for noise production in transformers are:

- 1] Magnetostriiction
- 2] The mechanical vibrations caused by the lamination
- 3] The mechanical vibration of the tank walls
- 4] Damping.

Worked Example On Parallel Operation of Transformers

Two single-phase transformers with equal turns have impedances of $(0.6 + j4) \Omega$ and $(0.8 + j10) \Omega$ with respect to the secondary. If they operate in parallel determine how they will share total load of 120 kW at 0.8 power factor lagging.

Soln

Impedance of transformer A, $Z_A = 0.6 + j4$

Impedance of transformer B, $Z_B = 0.8 + j10$

Total load to be shared = 120 kW at 0.8 p.f lagging

Find: Load shared by transformer A, P_A

Load shared by transformer B, P_B

Now $\cos \phi = 0.8$

$$\phi = -36.9^\circ$$

$$\text{Total load in kVA} = \frac{120}{0.8} = 150$$

Hence load to be shared, $P = 150 < -36.9 \text{ kVA}$. (21)

$$Z_A = (0.6 + j4) \Omega = 4.04 \angle 81.5^\circ \Omega$$

$$Z_B = (0.8 + j10) \Omega = 10.03 \angle 85.4^\circ \Omega$$

$$Z_A + Z_B = (0.6 + j4) + (0.8 + j10) =$$

$$= (1.4 + j14) = 14.07 \angle 84.3^\circ \Omega$$

Load shared by Transformer A.

$$P_A = P \times \frac{Z_B}{Z_A + Z_B} \text{ kVA}$$

$$= 150 < -36.9 \times \frac{10.03 \angle 85.4}{14.07 \angle 84.3}$$

$$= 106.93 < -35.8 \text{ kVA}$$

$$= 86.72 \text{ kW [i.e. } 106.93 \times 0.8 \text{]}]$$

Load shared by Transformer B

$$P_B = P \times \frac{Z_A}{Z_A + Z_B} \text{ kVA}$$

$$= 150 < -36.9 \times \frac{4.04 \angle 81.5}{14.07 \angle 84.3}$$

$$= 43.07 < -39.7 \text{ kVA}$$

$$= 33.14 \text{ kW [i.e. } 43.07 \times 0.8 \text{]}]$$

Check $P = P_A + P_B$

$$= 86.72 + 33.14 = 119.86 \text{ kW}$$

2] Two single phase transformers are connected in parallel on both primary and secondary. One transformer has a full-load resistance drop of 1% and reactance drop of 6%. For the other transformer the corresponding values are 1.5% and 5% respectively. How will the following loads be shared by the two transformers

Solution (i) 160 kW at 0.8 p.f. lagging

(ii) 155 kW at 0.9 p.f. leading

Solution

Impedance of transformer A, $Z_A = (1 + j6)$

Impedance of transformer B, $Z_B = (1.5 + j5)$

$$Z_A = (1 + j6) = 6.08 \angle 80.54^\circ$$

$$Z_B = (1.5 + j5) = 5.22 \angle 73.3^\circ$$

$$Z_A + Z_B = (1 + j6) + (1.5 + j5) = (2.5 + j11) \\ = 11.28 \angle 72.2^\circ$$

(i) 160 kW at 0.8 p.f. lagging

Total load in kW = 160 kW

$\cos \phi = 0.8$ lagging

$$= -36.9^\circ$$

$$\text{Total load in kVA} = \frac{160}{0.8} = 200 \text{ kVA}$$

Hence, total load to be shared, P

$$= 200 \angle -36.9^\circ$$

Load shared by transformer A

$$P_A = P \times \frac{Z_B}{Z_A + Z_B} = 200 \angle -36.9^\circ \times \frac{5.22 \angle 73.3^\circ}{11.28 \angle 72.2^\circ}$$

$$= 92.55 \angle -35.8^\circ$$

$$= 92.55 \angle -40.8^\circ \text{ kVA}$$

$$= 70.06 \text{ kW}$$

$$(92.55 \times 0.8 = 74.04)$$

$$74.04$$

Load shared by transformer B

(22)

$$P_B = P \times \frac{Z_A}{Z_A + Z_B} = 200 \angle -36.9^\circ \times \frac{6.08 \angle 80.54}{11.28 \angle 77.2}$$

$$= 107.8 \angle -33.56 \text{ kVA} = 89.83 \text{ kW}$$

⑪ 155 kW at 0.9 pf leading

Total load in kW = 155 kW

cos ϕ = 0.9 leading

$$\phi = 25.84^\circ$$

$$\text{Total load in kVA} = \frac{155}{0.9} = 172.2 \text{ kVA}$$

Hence total load to be shared, $P = 172.2 \angle 25.84$

Load shared by transformer A

$$P_A = P \times \frac{Z_B}{Z_A + Z_B} = 172.2 \angle 25.84^\circ \times \frac{5.22 \angle 73.3}{11.28 \angle 77.2}$$

$$= 79.69 \angle 21.94 \text{ kVA} = 73.9 \text{ kW}$$

Load shared by transformer B

$$P_B = 172.2 \angle 25.84 \times \frac{Z_A}{Z_A + Z_B}$$

$$= 172.2 \angle 25.84 \times \frac{6.08 \angle 80.54}{11.28 \angle 77.2}$$

$$= 92.82 \angle 29.2^\circ$$

$$= 87.1 \text{ kW}$$

N.B

If the impedance given is refer to different rating of transformer, these should be adjusted to the same base kVA

e.g. impedance of 500 kVA referred to base kVA of 1000 kVA

e.g.

$$Z_B = \text{impedance of } 500 \text{ kVA} \times \frac{1000 \text{ kVA}}{500 \text{ kVA}}$$

$$= (3 + j6) \times \frac{1000}{500} = (6 + j12)$$

